



PQLI

Guide Series

Product Quality Lifecycle Implementation (PQLI®)
from Concept to Continual Improvement

Part 4 – Process Performance and Product Quality Monitoring System

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Design Space

CQA

Pharmaceutical Quality System

Control Strategy

CPP

PQLI



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Lifecycle
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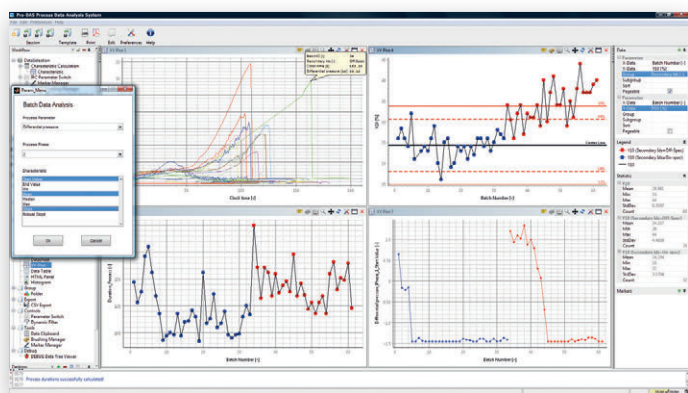


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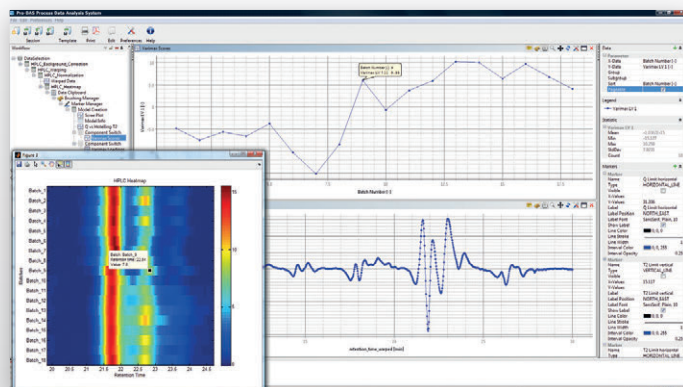
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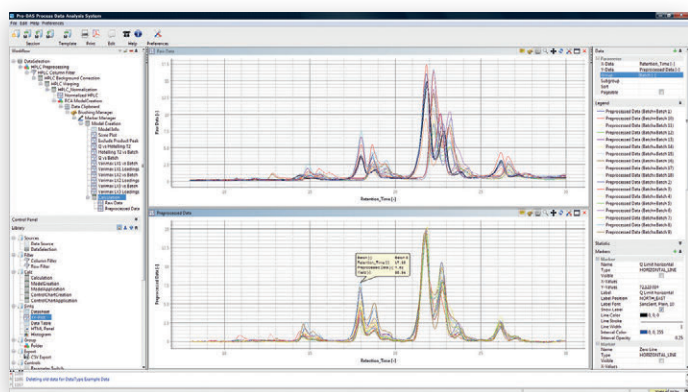
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PQLI

Guide Series

**Product Quality Lifecycle Implementation (PQLI®)
from Concept to Continual Improvement**

Part 4 – Process Performance and Product Quality Monitoring System

Disclaimer:

This Guide aims to provide “how to” guidance for establishing and implementing a Process Performance and Product Quality Monitoring System (PP&PQMS) in line with the expectations of the ICH Q10 guideline. ISPE cannot ensure and does not warrant that a system managed in accordance with this Guide will be acceptable to regulatory authorities. Further, this Guide does not replace the need for hiring professional engineers or technicians.

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Preface

The ISPE Product Quality Lifecycle Implementation (PQLI) Good Practice Guide (GPG): Process Performance and Product Quality Monitoring System (PP&PQMS) is Part 4 of the ISPE Guide Series: PQLI from Concept to Continual Improvement, which is a product of the ISPE PQLI program.

The modern pharmaceutical quality system described in ICH Q10 is a holistic approach which helps to facilitate the consistent development and production of high quality pharmaceutical products. The ISPE PQLI GPG: Process Performance and Product Quality Monitoring System considers process performance and product quality monitoring systems established to support products and processes developed using the enhanced, Quality by Design approach. This Guide is intended to be compatible with the process validation guidance issued by the FDA and the draft guideline on process validation issued by the EMA.

This Guide is relevant to the development and manufacture of drug substance and drug product for both small and large molecules. It is also applicable to new and existing products.

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1 Introduction

1.1 Objectives

The ICH Q10 guideline, Pharmaceutical Quality System [1], states that a process performance and product quality monitoring system (PP&PQMS) should be planned and executed to ensure that process performance and product quality are maintained in a state of control. The objective of this Guide is to provide “how to” guidance for establishing and implementing a PP&PQMS in line with the expectations of the ICH Q10 guideline. Examples of technical and scientific methodology and supporting management processes are provided.

Part 4 of the ISPE PQLI Guide Series, PP&PQMS, considers process performance and product quality monitoring systems established to support products and processes developed using the enhanced, Quality by Design (QbD) approach as discussed in ICH guidelines:

- ICH Q8(R2), Pharmaceutical Development [2]
- ICH Q9, Quality Risk Management [3]
- ICH Q11, Development and Manufacture of Drug Substances [4]

These systems are exemplified in other parts of the ISPE PQLI Guides Series including, Part 1, Product Realization using Quality by Design, Concepts, and Principles [5] and Part 2, Product Realization using Quality by Design (QbD), Illustrative Example [6].

This Guide is intended to be compatible with the process validation guidance issued by the FDA [7] and the draft guideline on process validation issued by the EMA [8].

This Guide should be read in conjunction with Part 3 of the ISPE PQLI Guide series, Change Management System [9]. These two Guides are PQS parts of the PQLI Guide series and support implementation of two of the Pharmaceutical Quality System (PQS) elements recognized in ICH Q10 [1] as enhancements of regional Good Manufacturing Practice (GMP) regulations.

1.2 Scope

This Guide (as is the case with ICH Q10 [1]) is relevant to the development and manufacture of drug substance and drug product for both small and large molecules. It is also applicable to new and existing products.

A PP&PQMS should apply to an entire organization or a site or division within an organization. The PP&PQMS should apply to all products within the assigned part of an organization. Each product should have its own control strategy and associated process performance and product quality monitoring program.

This Guide applies across all phases of the product lifecycle from development to product discontinuation. In the pharmaceutical development phase, data and knowledge emerge which support the development of the techniques and methodologies, including appropriate acceptance criteria which are monitored within the PP&PQMS. During the development phase, even in the later stages (e.g., Phase 3 clinical supply manufacture) application of the PP&PQMS is likely to be limited. However, in the technology transfer phase, there is potential to experiment with these techniques and methodologies. Full implementation will usually occur throughout the commercial manufacturing phase and will be gradually discontinued during the product discontinuation phase.¹

¹ Stability testing may still need to be performed.

The PP&PQMS techniques and methodologies suggested in this Guide are relevant to products developed using both enhanced, QbD approaches (as given in Appendix 1 of ICH Q8(R2) [2]) and to those products developed using the traditional approach.

The scope of these recommendations focuses on process performance and product quality data from manufacture of batches; therefore, this includes all product rejections, non-conformances, and deviations. Evaluation of external product quality data such as recalls, complaints, audits and regulatory inspections and findings as they impact the state of control is addressed briefly in Section 3.6 of this Guide. However, these are covered by other elements of the quality system, e.g., the Corrective Action and Preventative Action (CAPA) system and management review of process performance and product quality.

1.3 Benefits

An important benefit of developing and using the techniques and methodologies suggested in this Guide is that maximum opportunities should arise to gain the benefits listed in Annex 1 of ICH Q10 [1], which is reproduced below as Table 1.1.

Table 1.1: Potential Opportunities to Enhance Science- and Risk-Based Regulatory Approaches

Scenario	Potential Opportunity
Comply with GMPs	Compliance – status quo
Demonstrate effective pharmaceutical quality system, including effective use of quality risk management principles (e.g., ICH Q9 and ICH Q10)	Opportunity to: • Increase use of risk-based approaches for regulatory inspections
Demonstrate product and process understanding, including effective use of quality risk management principles (e.g., ICH Q8 and ICH Q9)	Opportunity to: • Facilitate science based pharmaceutical quality assessment • Enable innovative approaches to process validation • Establish real-time release mechanisms
Demonstrate effective pharmaceutical quality system and product and process understanding, including the use of quality risk management principles (e.g., ICH Q8, ICH Q9, and ICH Q10)	Opportunity to: • Increase use of risk-based approaches for regulatory inspections • Facilitate science-based pharmaceutical quality assessment • Optimize science- and risk- based post-approval change processes to maximize benefits from innovation and continual improvement • Enable innovative approaches to process validation • Establish real-time release mechanisms

Using the techniques and methodologies suggested in this Guide should lead to opportunities to gain many of the benefits listed and discussed in Part 1 of this Guide series [5], particularly benefits to:

- Improve manufacturing efficiency (e.g., process robustness)
- Support continual improvement
- Propose regulatory flexibility

1.4 Key Concepts

The Process Performance and Product Quality Monitoring System (PP&PQMS) should emerge in the later stages of product realization (ICH Q10 [1]). Operation of the PP&PQMS should contribute to providing opportunities for continual improvement and illustrates the utility of pharmaceutical quality management over the product lifecycle. A number of important considerations have gone into this Guide:

- The principles can be applied whether using a minimal or enhanced, QbD approach to development.
- In using enhanced, QbD approaches, this Guide is designed to make the most of the potential opportunities described in ICH Q10 Annex 1 [1].
- This Guide is not intended to provide a regulatory pathway. The focus is on technical underpinnings. Regulators in the various authorities around the world indicate requirements.
- This Guide will through implementation pathways and examples focus on the potential opportunities for continual improvement activities by availing the knowledge obtained over the product lifecycle.
- This Guide also provides for the opportunity and the responsibility of the manufacturer to be proactive in monitoring and detecting trends, and taking the necessary steps to produce products consistently of the intended quality.
- The principles of Quality Risk Management (QRM) and knowledge management will be highlighted as they support this PQS element, but they are not the main focus.
- Explanations are given to allow bridging between use of the term “specification” using ICH terminology and “specification” as it is often used in other guidance documents and engineering literature. This document uses the ICH definition as discussed in Section 1.5.1 of this Guide.

The Process Performance and Product Quality Monitoring System section introduces the role of Quality Steward as an owner of product quality. Responsibilities of the role are discussed and examples given in Section 4 of this Guide.

1.5 Definitions

1.5.1 Specification

Throughout this part of the Guide series, the term “specification” is used and some further explanation is required regarding how this term is used in this document and specifically in relation to a PP&PQMS. A specification (ICH Q6A [10] and ICH Q6B [11]) is defined as:

A list of tests, references to analytical procedures, and appropriate acceptance criteria which are numerical limits, ranges, or other criteria for the tests described. It establishes the set of criteria to which a drug substance or drug product should conform to be considered acceptable for its intended use. “Conformance to specifications” means that the drug substance and / or drug product, when tested according to the listed analytical procedures, will meet the listed acceptance criteria. Specifications are critical quality standards that are proposed and justified by the manufacturer and approved by regulatory authorities.

This definition specifically refers to drug substance and drug product. In the context of this document and a PP&PQMS, a specification also could apply to materials which are included in a control strategy and potentially a PP&PQMS, e.g., excipients as discussed in ICH M4Q(R1), Common Technical Document for Excipients (Section 3.2.P.4.1) [12]. ICH M4Q(R1) references ICH Q6A [10] and Q6B [11] when discussing excipient specifications; , therefore, an adaptation of ICH definition for drug substance and drug product specification is applied to a relevant input material used in a control strategy, such as drug substance starting materials, excipients, in process materials,

primary packaging materials. By analogy with ICH Q6A [10] and Q6B [11] definition, a specification for a material contributing to a control strategy is a list of tests, references to analytical procedures, and appropriate acceptance criteria, which are numerical limits, ranges or other criteria for the tests described. It establishes the set of criteria to which an attribute should conform as part of a control strategy.

Control of parameters in a control strategy is usually expressed as ranges of values.

In the multi-factorial situation a control strategy could include an algebraic formula or model linking values for an attribute to values of other parameters or attributes, which together ensure product quality. In such cases, acceptance criteria for attributes of materials or ranges of parameter values could be established in the control strategy in combination with use of the formula or model.

The definition of specification applied to drug substance and drug product still applies.

1.6 Guidance Structure

This Guide provides suggested techniques, methodologies, and management processes for the practical establishment, implementation, and operation of a PP&PQMS across the product lifecycle.

Section 2 discusses sources of variability in pharmaceutical processes and introduces measurement of variability and a short discussion on distribution of data. There is also a short discussion of sampling. This section is intended to assist readers who are not experts in statistics to understand sources of variability and measurement of variability, and sampling in pharmaceutical processes.

Section 3 Establishment and Application of a PP&PQMS expands on Section 3.2.1 of ICH Q10 [1]. Management processes are discussed throughout this section; however, organizations are developing a new role of Quality Steward and a final section is devoted to discussing this role and provides two examples.

2 Variability and Sampling

This section aims to explain some sources of variability and measurement of variability in pharmaceutical processes.

2.1 Variability

Variability may be inherent to a pharmaceutical process or may result from extrinsic factors (e.g., personnel or environment). Variability may occur in attributes or parameters and can arise from several different sources. Understanding variability depends on being able to identify contributing factors and to measure the variability of intrinsic factors.

Variability should be studied during pharmaceutical development and further understanding should be gained during technology transfer. The level of understanding of variability is likely to depend on both the level of resource applied to the study and the development approach taken, e.g.:

- Where an enhanced, QbD approach has been used during pharmaceutical development, it is likely that major contributors (i.e., attributes and parameters) to process variability would be identified and studied.
- Where a minimal approach is used during pharmaceutical development, there is a greater likelihood that there are poorly understood variables that could contribute to overall process variability. These variables could be identified by a PP&PQMS, if the variability is likely to lead to a product Critical Quality Attribute (CQA) failing to meet acceptance criteria.

2.1.1 Variability of Analytical Measurement

Variability of analytical measurement is a recognized contributor to overall process variability. Measurement systems and methods for collecting data on parameters and attributes should be suitable for their intended use, and appropriately installed and verified as fit for purpose. These measurement systems should be included in routine maintenance and calibration programs, as appropriate. They may be:

- In-line
- On-line
- At-line
- Off-line

The accuracy and precision of the measurements should be considered during data analysis and setting of acceptance criteria, operational limits, and control limits.

For further information on factors that could contribute to the accuracy and precision an analytical procedure, see Appendix 1. (This appendix also provides information on some of the considerations for process analyzers, process controllers, and data acquisition systems.)

Where trend monitoring is used, it is important to ensure that the analytical procedures are consistent, validated and providing the same capacity where ever used. For further information on analytical procedures which may be used in a PP&PQMS, see Appendix 3.

Where effective process performance evaluation is part of a PP&PQMS, it is dependent on measurement variability and the measurement variability needs to be understood.

2.1.2 Overall Process Variability

Risk assessment can be used to identify the impact of variability as part of an enhanced, QbD approach. Risk management exercises can be performed periodically throughout the product lifecycle, in order to manage continually identified risks.

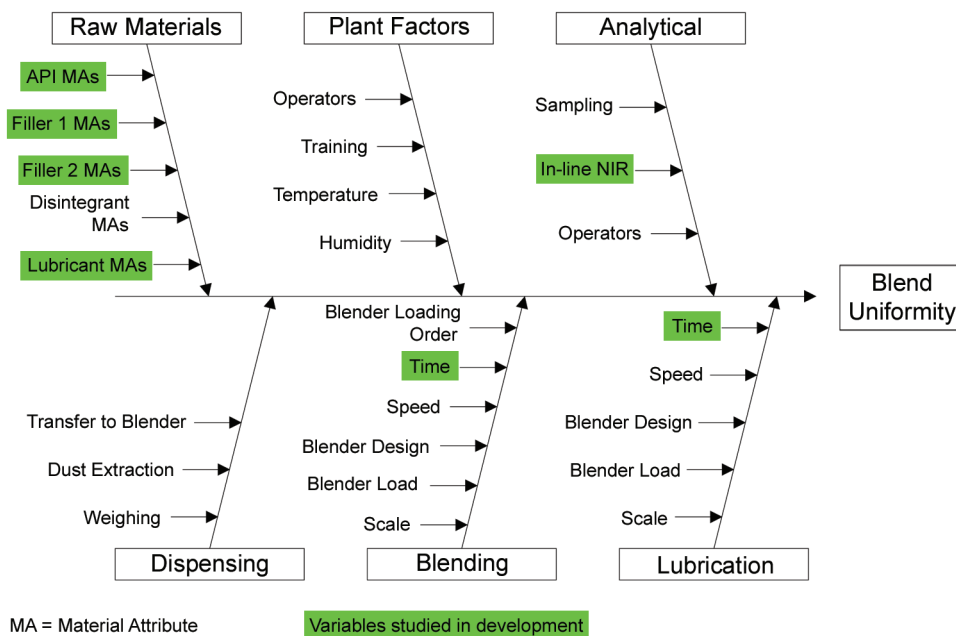
Visualization tools, such as an Ishikawa diagram (see Figure 2.1), may be used to identify potential risks (i.e., variability). Risk priority number charts from a Failure Mode and Effects Analysis (FMEA) and cause and effect matrices, which summarize the output from a risk assessment exercise, also may be used.

Figure 2.1 shows an example Ishikawa diagram for blending and lubrication unit operations for factors which can impact blend uniformity. Blend uniformity is an in-process quality attribute that contributes to the Uniformity of Dosage Units (UDUs) product CQA. The example Ishikawa diagram shows that there are residual lower risk factors which could contribute to overall process variability. A PP&PQMS could be used to confirm (using in-line near infrared (NIR)) that overall process variability remains in a state of control.

The risk management team should review the current level of product and process understanding including:

- Development studies
- Technology transfer activities
- Output from process validation programs

Figure 2.1: Example of an Ishikawa Diagram for Blending and Lubrication Unit Operations



Notes for Figure 2.1:

1. This example Ishikawa diagram is based on Figure 2.15 in Part 2 of this Guide series – Product Realization using Quality by Design: Illustrative Example [6]. (The names of the drug substance and excipients have been made generic.)
2. Factors studied during pharmaceutical development, which were originally considered the highest risk, have been identified by the green shading. Development studies reduced the risk of these factors to acceptable levels by setting appropriate acceptance criteria for inclusion in the control strategy.

3. The analytical procedure chosen for the unit operation (i.e., in-line NIR) is included.
4. Each branch of the diagram lists the risks, i.e., factors for which the variability may impact a CQA.
5. Each branch lists variables related to a single theme, e.g. analytical, raw materials, etc.

The risk management process usually uses a team of Subject Matter Experts (SMEs) from a variety of relevant disciplines. Risk management exercises should be performed to establish Critical Process Parameters (CPPs) and CQAs forming the basis of process validation programs (e.g., before and after a study to support commercialization of a process) and ongoing evaluation of process performance during the commercial manufacturing phase of the lifecycle,² i.e., the PP&PQMS. For further information on process validation, see the FDA Process Validation guidance [7].

Where possible, individuals involved early in the product lifecycle should be involved in risk management exercises later in the product lifecycle, to give continuity and provide knowledge. Risk management exercises also may be used to train new personnel and pass on process knowledge.

For further information on the application of the risk management process to the development of a formulation and associated process, see Section 2.2 of Part 2 of this Guide series – Product Realization using Quality by Design: Illustrative Example [6].

The residual level of risk in complying with drug substance or drug product CQA acceptance criteria should be assessed. (This may be considered as the level of risk of the control strategy failing.)

For further information on understanding and controlling variability in manufacturing processes, see Section 6.3, Continual Improvement Methodology Examples section of Part 2 of this Guide series – Product Realization using Quality by Design: Illustrative Example [6].

2.1.3 Distribution of Data

For the purposes of this Guide, it is assumed that data values follow a normal distribution.³ This distribution is based upon a measure of the mean value ($\bar{x}$) and the expected variation (standard deviation (s)) around the mean.

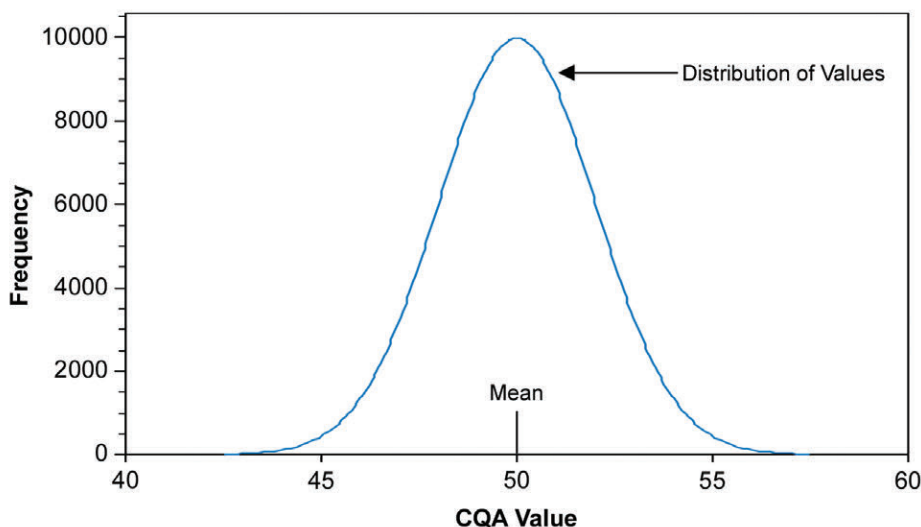
The distribution of data can apply to attributes or parameters for which a measured value is obtained. This type of data can be used to establish statistical process control limits for use in a PP&PQMS.

Note: for a non-normal data distribution expert opinion should be obtained.

Figure 2.2 provides an example of normally distributed data values for a single batch containing 500,000 items. The value of a hypothetical CQA is measured for every item within the batch, providing a 100% inspection. The bell curve shown represents the normal distribution of all 500,000 values obtained for the hypothetical CQA.

² Stage 2, Part 2, process performance qualification (and FDA Stage 3, continued process verification) of the FDA Process Validation guidance [7]).

³ For simplicity, the discussion in this Guide is limited to normally distributed processes with the caution that this assumption must hold true for a particular data set.

Figure 2.2: Distribution of Values of a Hypothetical Critical Quality Attribute**Notes for Figure 2.2:**Mean ($\bar{x}$) = 50Standard Deviation (s) = 2

Number of values = 500,000

As sample sizes for each batch need to be manageable, only a representative sample is usually taken rather than providing a 100% inspection. This means that a smaller number of values from the batch are sampled and some information is “lost” as a trade-off in order to obtain a manageable sample size.

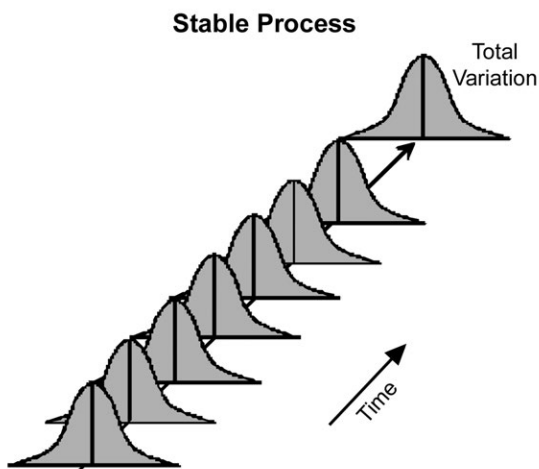
Where a smaller number of values from a batch are sampled, the mean and standard deviation for that batch can be estimated. These values for mean and standard deviation will probably be different to the values obtained if all values from the batch are measured. However, a range of values for mean and standard deviation for the several samples can be estimated with some confidence that the true mean and standard deviation values will not be statistically different. Larger sample sizes will give a more precise picture of the distribution of values.

A well-controlled process can generate multiple batches and a number of samples from each batch can be tested for a particular CQA; the results for these values for the CQA within each batch and for different batches can be plotted over time. These plots allow the trend in performance of the process to be observed. Figure 2.3 shows distribution curves for different batches over a period of time.

Common Cause Variation

For a stable process, individual samples can be combined to provide a picture of the total variation in the process. When the variation between batches is small, the total variation is similar to the variation within batches. In this case, the variability is often referred to as “common cause variation.”

Figure 2.3: Distribution Curves for Different Lots for a Stable Process



Graph taken from: "Quality Management Systems – Process Validation Guidance," The Global Harmonization Task Force, SG3/ N99-10:2004 (Edition 2) [13]

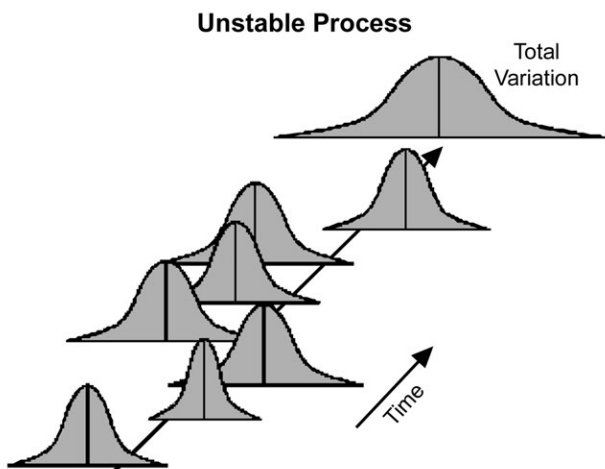
Special Cause Variation

Figure 2.4 shows an unstable process, where the effect of between batch variability on the total variation is much more pronounced. For an unstable process, the variability within one batch can be different to the variability in another batch, i.e.:

- The standard deviation for one batch can be different to the standard deviation for another batch.
- There are systematic differences in mean values between individual batches. The differences in mean values may be statistically significant.

The total variation (represented by the histogram at the top of the Figure 2.4) in the unstable process can only be seen retrospectively using all data. This type of variation may be referred to as "special cause variation." Common statistical tools such as the "t-test" and "F-test" can be used to help evaluate the type and degree of a special cause variation.

Figure 2.4: Distribution Curves for Different Lots for an Unstable Process



Graph taken from: "Quality Management Systems – Process Validation Guidance," The Global Harmonization Task Force, SG3/ N99-10:2004 (Edition 2) [13]

2.2 Sampling

A sample should be representative of a batch. Considerations when establishing sampling plans include:

- Clear definition of the objective of the sampling plan
- Definition of the sampling plan relevant to requirement, examples of which may include:
 - Random
 - Stratified
 - Composite
 - Systematic, e.g., beginning, middle and end of a unit operation
 - Targeted (e.g., worst case)
- Taking account of sampling and measurement error
- Determination of the relationship between sample size, statistical power, and statistical precision
- The type of change to the process and the size of change desired to be detected

Sampling plans are usually proposed during the technology transfer part of the lifecycle. Sampling plans are usually established to support a process validation program as part of the introduction of commercial manufacturing, e.g., Stage 2, process performance qualification of FDA Process Validation guidance [7]. A sampling plan may be modified to support Stage 3, continued process verification.

A sampling plan is an output from a risk management exercise for establishing a PP&PQMS. The sampling plan should be reviewed to ensure that it is representative of the batch.

For further information on sampling, see the US Government, Department of Defense standard, MIL-STD-1916 [14].

3 Establishment and Application of a Process Performance and Product Quality Monitoring System

3.1 Introduction

This section discusses considerations for establishing and applying a PP&PQMS. Topics are listed in the same order as in ICH Q10, section 3.2.1 sub sections (a) to (f) [1].

Table 3.1 provides a summary of considerations for establishing and applying a PP&PQMS.

Table 3.1: Application of Process Performance and Product Quality Monitoring System throughout the Product Lifecycle

	Pharmaceutical Development	Technology Transfer	Commercial Manufacturing	Product Discontinuation
ICH Q10, Table I	Process and product knowledge generated and process and product monitoring conducted throughout development can be used to establish a control strategy for manufacturing.	Monitoring during scale-up activities can provide a preliminary indication of process performance and the successful integration into manufacturing. Knowledge obtained during transfer and scale-up activities can be useful in further developing the control strategy.	A well-defined system for process performance and product quality monitoring should be applied to assure performance within a state of control and to identify improvement areas.	Once manufacturing ceases, monitoring such as stability testing, should continue to completion of the studies. Appropriate action on marketed product should continue to be executed according to regional regulations.
Key Objectives of a PP&PQMS	Understand CQAs, parameters and attributes to enable a control strategy to be proposed.	Modify and finalize control strategy. Select CQAs, parameters and attributes to evaluate during process performance qualification. Select CQAs, parameters and attributes for initial process performance and product quality monitoring.	Execute facility/ equipment qualification and process performance qualification. Confirm control strategy as part of process validation program. Evaluate data from initial process performance and product quality monitoring (FDA Process Validation, Stage 3, and continued process verification). Execute in a state of control. Opportunities for continual improvement, including modification of control strategy.	Continue to fulfill process performance and product quality monitoring required by regional regulations.

Table 3.1: Application of Process Performance and Product Quality Monitoring System throughout the Product Lifecycle (continued)

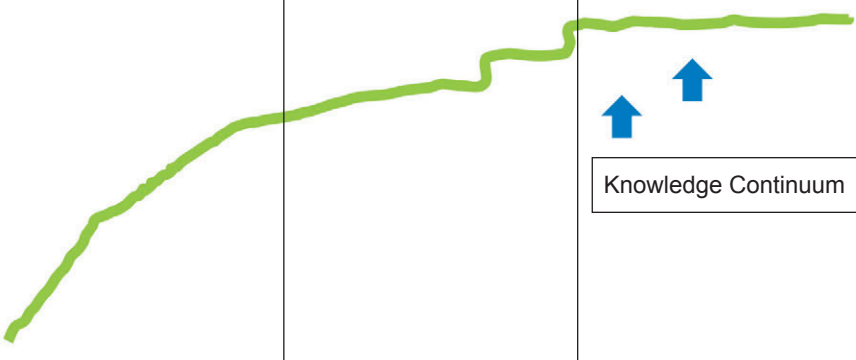
	Pharmaceutical Development	Technology Transfer	Commercial Manufacturing	Product Discontinuation
Product and Process Knowledge				
Techniques and methodologies for PP&PQMS	Develop Control Strategy: - Measurement Strategy Propose CQAs and CPPs and acceptance criteria. Start to understand variability by experimentation: - Multi-factorial experiments (DoEs) - Trial and error QbD Approach: - Risk management PAT Scale down Process modeling	Establish Control Strategy: - Further understanding of variability - Confirm CQAs and acceptance criteria - Confirm CPPs and ranges - Confirm other material attributes and acceptance criteria Process Model confirmation: - E.g., Monte Carlo simulation Quality risk management	Installation and Operational qualification: - Facilities - Utilities - Equipment - Analytical tools Ongoing evaluation of process performance and identification of opportunities for continual improvement: - Process variability reduction - Enhancing control strategy based on current knowledge of product and process Multivariate data analysis Process Capability Process Control: - Statistical Process Control	
Notes: - The top row (ICH Q10, Table I) is reproduced from ICH Q10 [1] and gives example applications of a PP&PQMS in the pharmaceutical lifecycle. - The second row (Major Activities for a PP&PQMS) summarizes major activities in the relevant part of the lifecycle.				

Table 3.1: Application of Process Performance and Product Quality Monitoring System throughout the Product Lifecycle (continued)

Notes: (continued)

- The third row (Product and Process Knowledge) shows how product and process knowledge may change over the lifecycle. “Step” increases in knowledge could be associated with implementation of continual improvement, the opportunity for which may arise from application of the PP&PQMS.
- The fourth row (Techniques and Methodologies for PP&PQMS) illustrates some of the technologies and methodologies which could be applied during development and application of a PP&PQMS.

Abbreviations:

CQA	Critical Quality Attribute
DoE	Design of Experiments
PAT	Process Analytical Technology

3.2 Establishment and Implementation of a Control Strategy as a Basis for a PP&PQMS

A control strategy serves as the basis for establishing PP&PQMS requirements for a product.

It is helpful to understand the different tools and activity stages that can be utilized throughout the process of implementing a control strategy that will be the foundation of the PP&PQMS for a given product.

A control strategy should be developed during the pharmaceutical development stage, modified and set during the technology transfer phase, and confirmed as part of the process validation program in the early phase of commercial manufacturing, e.g., as part of FDA Stage 2 – process qualification. The on-going effectiveness of the control strategy should be evaluated and continually improved during the commercial manufacturing phase of the lifecycle (see Stage 3, continued process verification of the FDA process validation guidance [7]).

A multi-disciplinary risk management exercise should be performed for each product. This exercise is part of assessing which aspects of the control strategy should be included in the scope of the PP&PQMS.

Establishing and implementing a control strategy is considered an important foundation for the planning and execution of a PP&PQMS. For further information, see Appendix 2.

When real time release testing is included as part of a control strategy, assurance of conformance of a batch with CQA acceptance criteria should be equivalent or better than when finished product testing is applied.

Sampling plans should be re-considered following the early phases of the process validation program as discussed, see Section 2.2 of this Guide.

For further information on feed forward and feedback control, see Juran’s Quality Handbook [15] and Process Dynamics, Modeling, and Control [16].

3.3 Provide the Tools for Measurement and Analysis of Parameters and Attributes

Factors potentially contributing to overall process variability are discussed in Section 2.1 of this Guide. Key outputs from the team establishing the PP&PQMS requirements for a product based on an already existing control strategy are:

- Selection of attributes and parameters to be included:

- The highest risk drug product CQAs (as determined from the risk analysis of all work performed to that point) will be included. In a real time release testing scheme, where the drug product CQA may not be measured directly on end product, appropriate attributes, parameters, and model predictions which provide assurance that the drug product CQA meets acceptance criteria should be included in the PP&PQMS.
- Confirmation that analytical procedures are in a state of control
- Statistical tools to be used, e.g., control charts, histograms
- Draft of rules for monitoring and trending

Only after sufficient data points (e.g., batches from a stable process) have been obtained is it possible to set meaningful statistical process control limits (typically a minimum of 25 data points or batches). These limits should be revised periodically as more data become available, which should characterize better the variation of the current, typical operation. “Temporary” process control limits, based on historical data, prior experience, and process understanding can be used until a sufficient amount of data is available. For products with relatively low frequency of manufacture (e.g., four or less batches per year), it may not be possible to set statistically meaningful process control limits; therefore, it may be necessary to retain process control limits using empirical methods.

When sufficient data have been generated, the implementation team and management should consider:

- Calculation of baseline performance to provide statistical process control limits and process capability measurements
- Establishment of rules for monitoring and trending
- Defining Out of Trend (OOT) limits and establishment of a process for managing OOT results
- Formalizing investigations and communication required for OOT investigations
- The process for updating baseline calculations based on amount of data generated, process change implementation, etc.

Data should be collected within the Pharmaceutical Quality System (PQS) for trending and monitoring. Appropriate management review of the program and outputs should be performed to ensure that any process performance or product quality issues are dealt with in a timely manner. The purpose of trending is to identify both negative and positive trends.

- Identifying a negative trend before it affects product quality can guard against manufacturing of unacceptable material. Investigations of negative trends and Corrective Actions and Preventative Actions (CAPAs) can facilitate process improvement with the measurable goal of reducing variability.
- Identification of positive trends where variability has been reduced provides a second opportunity to capture and reproduce an improvement, as well as demonstrate the effectiveness of CAPAs or continual process improvements.

The immense volume of data available in modern manufacturing requires strategic planning by people with knowledge of the requirements of pharmaceutical data integrity and storage and the use of information technology where available and appropriate.

Therefore, careful consideration of the strategy and implementation of relevant and practical statistical tools for the monitoring program is of fundamental importance.

Hence as discussed above, elements to be considered in the implementation of trending programs **before implementation** include:

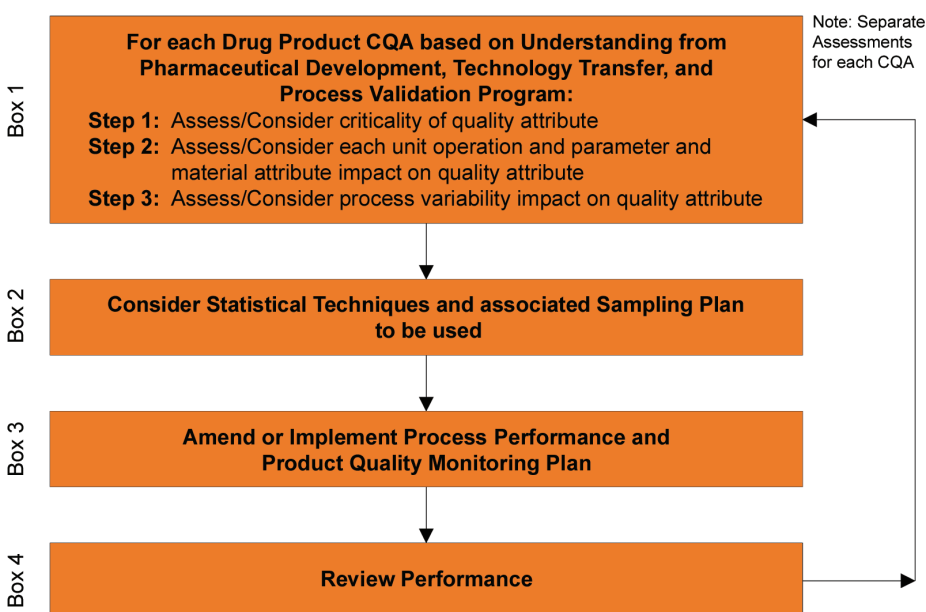
- What is to be trended?
- What are the triggers for action?
- What is the resolution of an actionable event?

The danger is that when misapplied these tools have as much potential to contribute to the confusion as they do to solve it. The approach in ICH guidelines ICH Q8(R2) [2], ICH Q9 [3], ICH Q10 [1], and ICH Q11 [4] begins with assessment of risk to the patient and prioritization of work. ICH Q10 (section 3.2.4) [1] indicates that trending programs should be evaluated by management and appropriate actions should be identified, such as improvements to the process made throughout the product lifecycle.

When implementing a PP&PQMS across the lifecycle it should be recognized that data generated by experimental design are fundamentally different from data collected by the routine monitoring of operations through in-process checks and acceptance sampling. Data from routine monitoring of operations are typically generated by relatively stable processes where variation in the inputs is minimized by the control strategy and varied by chance. Experimentation, on the other hand, deliberately changes inputs to measure the effect on the outputs.

A simple flow chart for implementing a PP&PQMS is given in Figure 3.1.

Figure 3.1: Implementation of a PP&PQMS



For Box 1, much of the information to assess/consider CQAs will have been conducted and finalized as part of pharmaceutical development, technology transfer, and marketing authorization approval. The team implementing the process performance and product quality monitoring plan should, in effect, perform a risk management exercise to determine residual risk (e.g., sources of variability not studied prospectively). The findings from this risk management exercise should be combined with consideration of appropriate statistical tools to be applied. (Appropriate statistical tools may already be in place as part of Stage 3, continued process verification of the FDA process validation guidelines [7]).

Process performance and product quality data should be evaluated in a timely and effective way based on manufacturing frequency and level of risk to facilitate the communication and escalation process to raise appropriate quality issues to senior levels of management for review. This evaluation may be combined with an annual product review (see Section 3.6 of this Guide). However, a more frequent periodic review may be appropriate to ensure a state of control and detection of near term adverse trends. It also may provide an evaluation of other quality indicator data, e.g., Out of Specification (OOS), complaints, and increased deviations (see Section 3.6.2 of this Guide).

A variety of statistical tools can be employed, e.g.:

- Simple descriptive statistics
- Statistical process control
- Process capability measurements

Note: it is recommended that practitioners consult the literature and arrange for appropriate statistical expertise to be on or available to the team implementing a PP&PQMS.

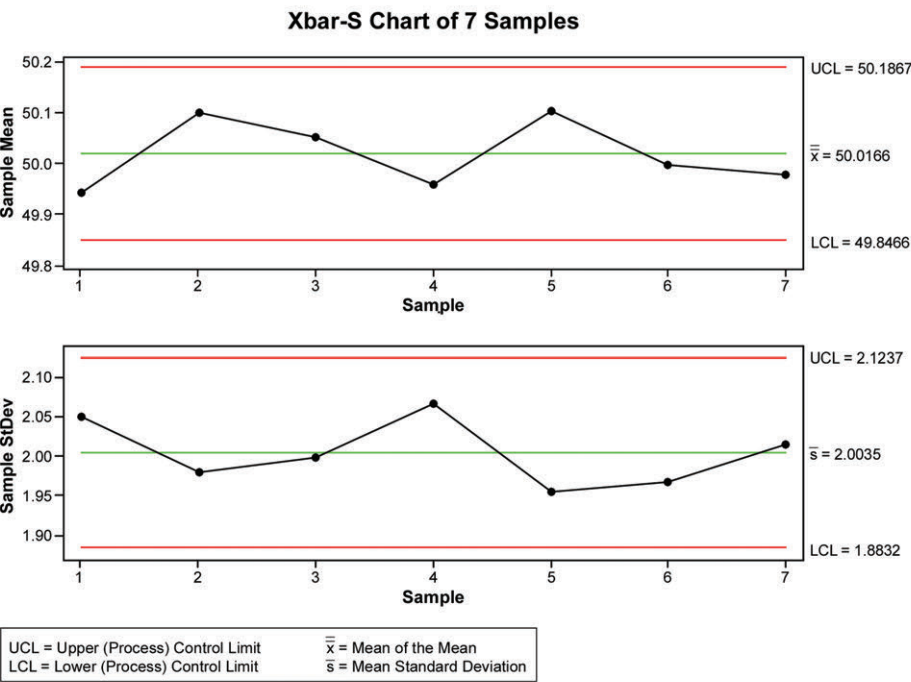
3.3.1 Simple Descriptive Statistics

Data could be presented in several ways, e.g.:

- Time series plots
- Histograms
- Box plots

Figures 3.2, 3.3, and 3.4 are examples. Figure 3.2 provides an example time series plot which includes statistical process control limits.

Figure 3.2: Summary Statistical Values from Samples from Different Lots from a Stable Process



Notes for Figure 3.2:

1. The upper chart shows 7 mean values, the mean of the means ($\bar{\bar{x}}$), Upper (Process) Control Limit (UCL) and the Lower (Process) Control Limit (LCL) for the mean of the means.
2. The lower chart shows the standard deviations (s) associated with these 7 mean values, showing the UCL, LCL, and mean standard deviation ($\bar{s}$).

(See Section 3.3.2 of this Guide for an explanation of how UCL and LCL values are set.)

For further information on box plots, please refer to the relevant literature, e.g., in Juran's Quality Handbook [15].

3.3.2 Statistical Process Control

This section discusses concepts of establishing a baseline with appropriate process control limits and setting rules for monitoring and trend analysis.

There are several types of statistical process control charts for a variety of different purposes, e.g.:

- Individual – moving range chart
- Average-standard deviation chart

For further information on recommendations for different types of charts, see ASTM E2587 Standard Practice for Use of Control Charts in Statistical Process Control [17].

For further information on techniques for statistical process control, including discussions of management philosophy and implementation, see Walter A. Shewhart [18], W.E. Deming [19], and Joseph M. Juran [15].

Establishing Baseline with Statistical Process Control Limits

Statistical process control limits reflect the boundary of typical operation. Typically, these limits are based on statistical criteria or may be based on an in-depth knowledge of a unit operation in order to detect changes in process performance. They are intended to detect unexpected changes in the manufacturing process that are not due to the natural variation within the process.

The data set used is considered critical to setting process boundaries. Process monitoring should start by collecting data and plotting it against time. Process monitoring should start before setting limits for acceptability of process performance. Once a sufficient body of data has been collected and any identified special cause variation addressed, it should be possible to consider calculating the boundaries on the expected performance of the process.

Figure 3.3 shows two charts which illustrate statistical process control limits by plotting the summary statistics of data with the probability limits of the typical process performance:

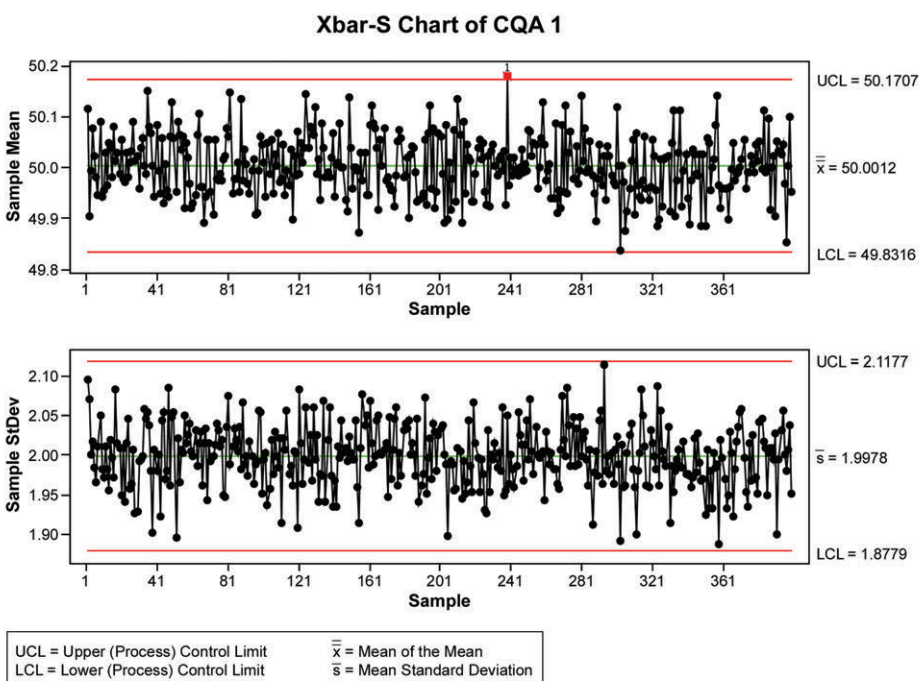
- In the upper chart, each dot represents the mean ($\bar{x}$) of a sample of n units.
- The lower chart shows one dot for the standard deviation of each sample of n units.
- The green line represents the typical value for the process and the red lines represent the probability limits that (in theory) encompass the samples 99.73% of the time when ± 3 standard deviations are used on a large, normally distributed data set:⁴

⁴ Note: although ± 3 standard deviations is valuable as a rough rule of thumb, other statistical tools exist to set statistical process control limits.

Note: statistical control limits are based on process performance and are not directly related to risk to the patient. They should be used to detect and control changes in the process. Statistical process control limits are specific to a process and should fall within the acceptance criteria or ranges which are included in a drug product, drug substance, or material specification included in a PP&PMQS monitoring plan/control strategy. They should not be confused with specification acceptance criteria which are more relevant to therapeutic or toxicological effect.

In Figure 3.3, a chart of values from 400 lots is plotted. The majority of values fall within the limits; however, one value of $\bar{x}$ (in red) is beyond the probability boundary of the Upper (process) Control Limit (UCL). This higher than expected variability may be indicating a special event, or it may be a false alarm, i.e., due to random patterns alone (an issue inherent to control charts). Statistical techniques exist for determining an average run length and the probability of a false alarm occurring. This and other tools should be considered as part of an evaluation involving a review of the statistical techniques applied and of an understanding of the process to determine whether this extreme value is to be expected or indicative of a problem in the process control requiring corrective or preventative action. (For further information, see Section 3.5 of this Guide.)

Figure 3.3: Summary Statistics for 400 Lots



Differences between the intra batch and inter batch variability should be considered when setting appropriate statistical process control limits. For further information, see ASTM E2587 Standard Practice for Use of Control Charts in Statistical Process Control [17].

With regards to variability, statistical process control limits test for changes to the:

1. Mean
2. Standard deviation

For further information on how to handle variability of standard deviations and the impact on process capability, see ASTM E2709 [20].

In the first phase of setting up a monitoring program, data should be trended without process control limits, until the process is stable and a sufficient body of data is accumulated to support calculations of meaningful statistical process control limits. In the subsequent phase, the process can be monitored and improvements identified, which may include decreasing special or common cause variability. Once the process is both stable and sufficiently capable (see Section 3.3.3 of this Guide), it is possible to switch to other monitoring approaches, such as cumulative sum control (CUSUM) charts [21] or exponentially-weighted moving average (EWMA) charts [21]. For processes with relatively small runs of material, other monitoring approaches may be considered. The appropriate tool should be chosen with support of statistical experts.

In addition to establishing the statistical process control limits, Alert and Action limits also can be established to support early detection of special cause variation events. Definitions of these limits may be:

- Alert limit – a limit reached prior to an action limit which, if exceeded, provides early warning of a potential change from normal operating conditions
- Action limit – a limit which, if exceeded, indicates a significant change from normal operating conditions

The difference between acceptance criteria or ranges, and statistical process control limits should be understood, because all processes, at times, may operate outside of statistical process control limits due to random patterns alone. Acceptance criteria or ranges are usually included in a drug product, drug substance, or material specification included in a PP&PMQS monitoring plan/control strategy.

Operation within approved CPPs, material attributes, and product specification acceptance criteria is a regulatory requirement. If a value from a process is outside a statistical process control limit, but remains within approved critical process parameter range or material attribute and product specification acceptance criteria, it may still be acceptable product. An investigation of historical data would be required to determine whether the value were a false alarm or a real issue, due to a special cause. The level of effort, formality, and documentation of an investigation into a result outside a statistical process control limit should be commensurate with the risk to product quality. Outcome of the investigation would not be grounds for rejection of a batch in and of itself.

Statistical process control should ensure ongoing consistency and predictable product quality by identifying changes to the process before they lead to values outside acceptance criteria/ranges approved in marketing authorizations.

Setting Baseline and Trend Analysis and Establishment of Rules for Monitoring and Trend Analysis

Data from, e.g., process control charts, can be evaluated using consensus Western Electric Company rules [22] to identify non-random patterns in the data.

Interpreting statistical rules, such as the Western Electric Company rules, should be carefully considered. For further information on the management process associated with dealing with values from monitoring programs which fall outside a statistical process control limit see Sections 3.4.1, 3.5.3, 3.6, and 4 of this Guide.

3.3.3 Process Capability

The goal of a PP&PQMS is to confirm that process performance and product quality are in a state of control. Measures of process stability and capability help in determining that a process is in a state of control. If a process is not sufficiently stable and/or capable a PP&PQMS should provide data to help identify sources of variation, which could lead to continual improvement opportunities.

A matrix of descriptions of stability and capability is given in Figure 3.4.

Figure 3.4: Matrix of Stable and Capable Processes

	Stable	Not Stable
Capable	The place to be!! Variation within and between batches acceptable	Variation between batches too high
Not Capable	Target not around center of specification limits and/or variation too large	Variation between batches too high and total variation too high

Process Stability

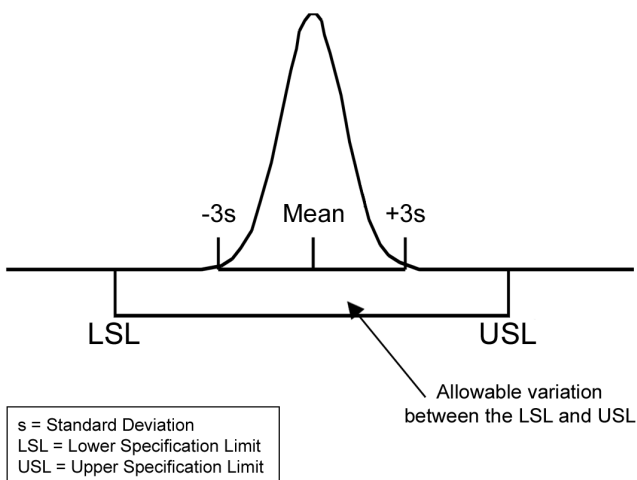
A manufacturing process that is stable ensures that the process delivers reliable results (i.e., consistent variability) over time and across different manufacturing sites, where applicable.

A well-designed and well-performing process should meet statistical process control limits and acceptance criteria given in a specification, or ranges given in a marketing authorization.

Process Capability

The total variation of measurement of a CQA is used to estimate the Process Capability for that CQA.

Figure 3.5 is a plot of the total variation of a CQA, e.g., of data from a process, with the mean and standard deviation (s) of +/- 3, and the specification acceptance criteria (lower specification limit, upper specification limit) indicated.

Figure 3.5: Process Capability Showing Relationship of Total Variation to Specification Limits for a Stable and Capable Process

Dividing allowable Specification Range by the Total Variation gives an indication of how well the process runs within specification acceptance criteria. This ratio is called the Process Capability Index, Cp, and is expressed as:

$$C_p = (USL - LSL) / 6s$$

Cp is a measure of capability. It is inversely proportional to the standard deviation; however, it does not consider whether the mean of the data is on target, i.e., at the center of the upper and lower specification limits.

Another capability measure is Cpk which measures distance between the mean value and closest specification limit, while taking account of the standard deviation. Cpk is expressed as:

$$Cpk = \min (USL - \bar{x}, \bar{x} - LSL) / 3s$$

The larger the index, the more capable a process is of meeting specification limits. Generally, a value of Cpk greater than 1.33 is considered desirable:

- Cpk values > 1.33 indicate a process corresponding to where 99.99% of reported values are expected to be within specification limits.
- Cpk < 1.33, normal variability of the process can produce products that fall outside specification limits
- Cpk < 1, the variability of the process is greater than the specification limits and can produce products that fall outside specification limits.

Cpk does not necessarily show if the process is on target, only the chance that values will fall outside specification limits.

The confidence interval associated with the capability indices also should be considered. For further information, see ASTM E2281-08a “Standard Practice for Process and Measurement Capability Indices” [23].

Cc measures the distance between actual mean and the target mean. On its own, Cc does not take account of the variation (standard deviation, s) in the process data.

$$Cc = \max (\bar{x} - \text{Target} / USL - \text{Target}, \text{Target} - \bar{x} / \text{Target} - LSL)$$

Cc is a measure of quality as it indicates whether a process is on target and likely to produce good yield. A value of zero for Cc indicates the process to be on target.

Although process capability demonstrates the historical performance of a process, it does not express process robustness. Process capability demonstrates only that the control strategy is effective; with total variation being within specification limits.

On their own process capability measurements are insufficient to justify removing of a control strategy element. Process capability measurements can be sufficient to help to justify changing between a normal, relaxed, or heightened level of monitoring. Precedence for this re-evaluation of the control strategy already exists; see the switching rules of the American National Standard Institute (ANSI) Z1.4 and Z1.9. [24].

Note: switching rules require careful monitoring and control. Organizations may find a statistically appropriate sampling plan designed specifically for their process to be more easily implemented and beneficial.

3.4 Analysis of Parameters and Attributes

The technical methods used for the analysis of data should be considered. For further information, see Section 3.3. of this Guide.

3.4.1 Management Review Process

A well thought out management process should be established to support the PP&PQMS. The management process should consider questions such as:

- Who has what responsibility to collect, evaluate, report, and review process performance and product quality monitoring data?
- Who can decide what actions can be taken if there are OOT events, e.g., to identify sources of variation and their root causes?
- How are decisions made to initiate, agree, and implement a continual improvement project which could involve many technical and regulatory disciplines within an organization, as well as consume significant resources?

Organizations may establish quality steward roles to help manage these questions. For further information, see Section 4 of this Guide.

Although the PP&PQMS is part of the quality system, the responsibility for executing different tasks may not be within the Quality Unit.

3.5 Identify Sources of Variation

The main sources of variation should be identified during pharmaceutical development. Application of statistical tools to routine process data can reveal additional sources of variation, which could be from, e.g., operating conditions. Any unacceptable level of variation should lead to an investigation. This may include the use of statistical tools and other techniques to identify the root cause of the variation.

Where it is impractical to remove the sources of variation, it may be possible to compensate for the variation by adjusting inputs. Engineering Process Control (EPC) uses real time monitoring of the output of a process to feedback changes to the inputs (feedback control). Changes to the inputs can be manual or automated, and are typically:

- Proportional to the measured output
- Integrative to the measured output
- Derivative to the measured output

Strategies also may include probabilistic process control and fuzzy logic control. For further information, see the Process Dynamics, Modeling, and Control [16], ISPE GAMP® 5 [25], the ISPE GAMP® GPG: A Risk-Based Approach to GxP Process Control Systems [26], and the Process Analytical Technology Guide [27].

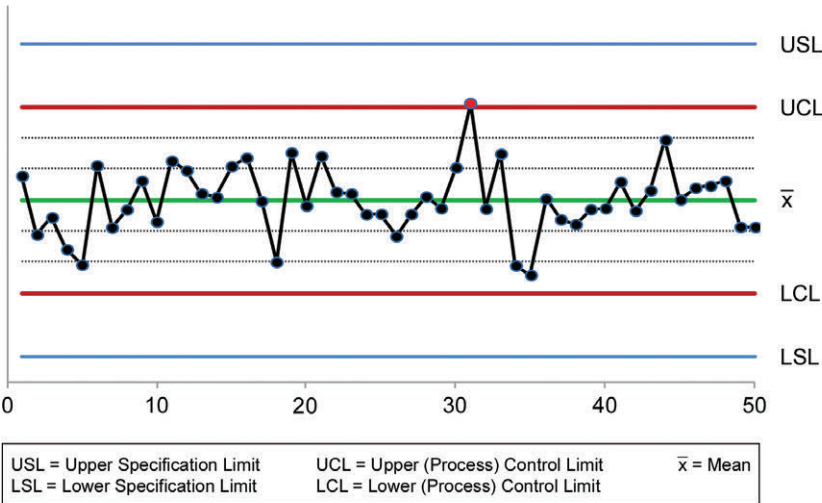
Organizations may establish quality steward roles to help manage the technical and operational complexity of process performance and product data generated across the product lifecycle as well as across products and locations. For further information, see Section 4 of this Guide.

3.5.1 Example of Application of Different Out of Trend Considerations

All statistical trends values may not have the same quality concern and the impact of identified statistical trends on product quality should be considered. An out-of-trend policy should be established, in order to use statistical process control and associated tests appropriately to monitor and improve product quality. The out-of-trend policy should facilitate appropriate actions commensurate with the degree of quality concern.

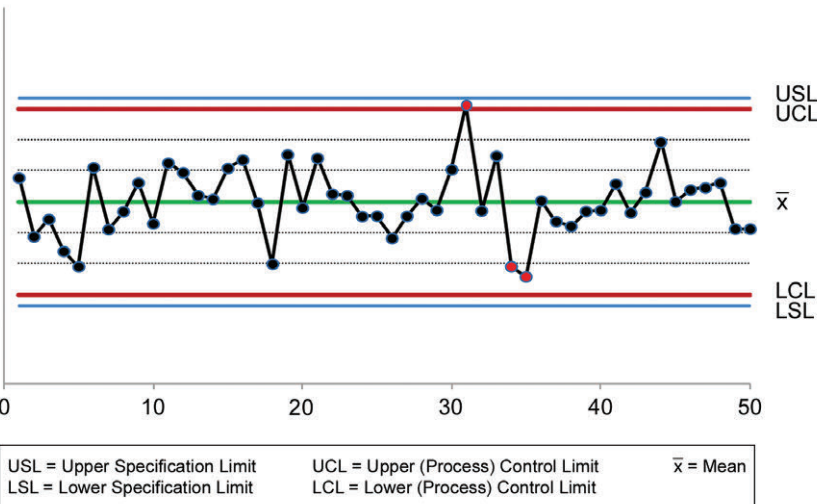
Figure 3.6 shows a Statistical Process Control (SPC) chart of a stable process. This process is highly capable, as the voice of the process (defined by the lower and upper process control limits) is well within the voice of the customer (defined by the lower and upper specification limits). In this case, a reasonable out-of-trend event may be limited to the data points that are outside statistical process control limits, e.g., the single red data point. When a data point exceeds the process control limits, actions should be taken to assess the potential special causes that led to the event.

Figure 3.6: Out of Trend Monitoring of a Stable and Highly Capable Process



In Figure 3.7, the same stable process is much less capable at meeting the specification limits, as the variation; therefore, the process control limits are much closer to the specification limits. In this case, reasonable out-of-trend criteria may need to be more sensitive and include additional trends/shifts, in addition to the use of process control limits. For example, an out-of-trend event may be triggered when two out of three consecutive data points are more than two standard deviations away from the process mean. In this case, there are three red data points which would be considered “out of trend.” The system facilitates proactive actions against potential disturbances when the process has a higher risk, by making the out-of-trend criteria more sensitive.

Figure 3.7: Out of Trend Monitoring of a Stable and Less Capable Process



3.5.2 Example of Use of Statistical Tools

Figure 3.8 shows a well-controlled commercial process that is under routine monitoring.

A CQA from 50 batches is trended. (The data for the 50 batches is shown as dark circles as an I-chart.) No special cause events are observed. A supplier of a key excipient contacts the manufacturer to explain that the site of manufacture of the excipient is being changed. The excipient from the new site meets all quality requirements.

The Cpk of the process, as determined from measurement of one CQA before change of excipient manufacturing site was around 1.3. It was decided from a process performance perspective that a process mean reduction of more than one standard deviation was not acceptable (i.e., the Cpk would be less than 1.0, while keeping the same variability). It was calculated that 11 batches should be monitored using excipient batches from the new manufacturing site to detect this unwanted shift with 90% power.

Figure 3.8 shows the data from the 11 new batches as open circles. Each new data point was compared to the process control limits (red lines) established by the pre-change process. The trends/shifts of the process were assessed. An out-of-trend event was observed (two out of three batches were more than two standard deviations away from the mean). This is shown in Figure 3.9 as a red circle in the I-chart. This out-of-trend event signaled a potential process shift due to the new manufacturing site of excipient. A t-test also confirmed that the new data set had a lower process mean.

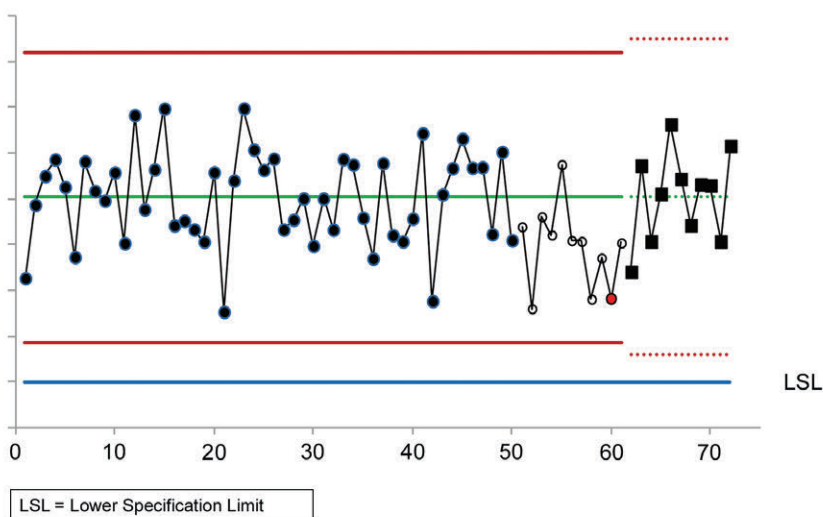
Although still meeting its specification requirements, the subtle excipient property change has had a negative impact on product quality. The design space derived from development studies shows that this product attribute is impacted by both the excipient quality and the compression operation. Adjusting the compression unit operation parameters, therefore, may compensate for the impact of the excipient property change. After calculation and implementation of the new compression conditions, a further 11 batches are manufactured (shown as dark squares in Figure 3.8).

Adjustment to the compression operation appears to bring the attribute back to the pre-change performance, as far as variation about the mean is concerned. The SPC chart baseline should be re-established due to the changes to the excipient and compression operation.

The data collected should be trended first as a time series, without “locked” process control limits. Once sufficient batches (e.g., ≥ 25) manufactured using the new condition, special events, if any, should be assessed. Updated process control limits also can be established for routine monitoring. These are shown by the dotted red line in Figure 3.8.

Through statistical monitoring and proactive adjustments, the manufacturer is able to control the attribute effectively. The product quality is not compromised by the changes and the process remains in a state of control.

Figure 3.8: SPC for a Process which Remains in a State of Control using Statistical Tools



3.5.3 Management Issues Involved in Identifying Sources of Variation

Organizations should create systems for managing values which are considered OOT, e.g., values which fall outside proposed Alert or Action limits. If a value falls outside an Alert limit, the following questions may be asked:

- Would a Product Quality Steward, if appointed, be informed immediately?
- What roles would local manufacturing and Quality Unit have?

The role of a Product Quality Steward also should be considered where a value is observed to fall outside the Action limit.

The management process should define how any personnel observing an OOT observation should deal with that event. In commercial manufacturing:

- The first personnel to review a process for a trend are likely to be the personnel who are able to prevent that process from going out of statistical process control
- These personnel also may be responsible for initiating an investigation when a trend indicates sufficient evidence that the process has changed

For further information, see Section 3.4.1 of this Guide.

Identification of a special cause variation and addressing and correcting potential causes through preventative actions should prevent a process from drifting out of statistical control. The first response to an OOT flag should be to review the historical data to determine the practical significance of the observed trend. This may be based on knowledge of a prior process drift with an assignable cause, inherent campaign-to-campaign variability, analytical method precision, etc. Some general rules also may be applied before investigations are triggered, e.g., multiple flags before initiating an investigation to avoid false alarms.

Example

A process is running in two commercial manufacturing sites in a global organization. The process technician at a site responsible for monitoring the dissolved oxygen of a bioreactor requires tools both to:

- See the trend in data
- Feedback into the system any assessment

It also may be beneficial for the process technician to be able to tune process parameters within approved ranges before the process goes out of Statistical Process Control (SPC).

At the analytical laboratory where each lot of Active Pharmaceutical Ingredient (API) grown in the reactor is tested for CQAs after manufacture, the technician performing the analysis also may be authorized to check the result for trends and escalate any potential issues at that site. This could be termed Statistical Quality Control (SQC).

A Quality Steward (see Section 4 of this Guide) at one site may then combine the data between one site and another site to check for trends. The data may then be refined into metrics for each manufacturing site.

At the division level the management process is repeated by combining data from each manufacturing site and refining information for each division. Ideally, this process continues all the way to senior management review (see Sections 3.6 and 4 of this Guide).

Metrics should be prioritized, summarized, and transformed into knowledge useful to the next level of review, as it may not be possible to review effectively all the data available from the preceding level in the organization.

Western Electric Company rules [22] tests may be useful to appropriately trained engineers and applied statisticians when troubleshooting on a manufacturing floor. However, applying all tests on control charts at every level of an organization may be costly, and has the potential to create confusion and undermine the goals of:

- Reducing variability
- Increasing quality
- Decreasing costs
- Decreasing risk to the patient

3.6 Include Feedback on Product Quality from both Internal and External Sources

The purpose of the PP&PQMS is to ensure that a state of control is maintained.

In addition to process performance and product release monitoring, feedback on product quality can arise from several sources, such as external sources, e.g.:

- Complaints
- Recalls
- Audits
- Regulatory inspections and findings

Sources of internal feedback include management review of product quality data.

All feedback sources, together with process performance and product release monitoring data, should feed into an organization's CAPA system. The PP&PQMSs and CAPA systems should be closely linked and may be a single management system. Organizations may have introduced quality steward roles to help manage the complexity of both PP&PQMSs and CAPA systems. See Section 4 of this Guide.

Suitable management processes should be established for obtaining, reviewing, and acting on product quality information arising from wide-ranging sources of variation (see Sections 3.4.1 and 3.5.3 of this Guide). Management processes and product quality stewardship role (see Section 4 of this Guide) should help with managing these complex information flows across the business.

3.6.1 Product Review

A product review process can be used to obtain feedback on product quality arising from many sources. A product review process may be established to meet GMP and regulatory requirements or it could be more extensive, as discussed in Section 3.2.4 of ICH Q 10 [1].

Product or quality review for marketed products is a GMP requirement from US, EU, and other regulatory agencies. It offers a basic platform for performing and documenting such reviews. Annual review is a minimum requirement. This review is separate from other reviews triggered by CAPAs or the need to understand intra-batch variability. A primary objective of the Annual Product Review (APR) is to verify the state of control and make recommendations for changes, as necessary. Integrating the documenting and reporting activities from the process performance and product quality monitoring programs into the APR process may ensure that evaluation of process performance and product quality is done formally.

Section 3.2.4 of ICH Q10 [1] states that information gathered from process and product monitoring should be reviewed by management. The description given in ICH Q10 [1] may be considered "the ideal" and as such the text is:

Management review should provide assurance that process performance and product quality are managed over the lifecycle. Depending on the size and complexity of the company, management review can be a series of reviews at various levels of management and should include a timely and effective communication and escalation process to raise appropriate quality issues to senior levels of management for review.

- (a) *The management review system should include:*
- (1) *The results of regulatory inspections and findings, audits and other assessments, and commitments made to regulatory authorities*
 - (2) *Periodic quality reviews that can include:*
 - (i) *Measures of customer satisfaction such as product quality complaints and recalls*
 - (ii) *Conclusions of process performance and product quality monitoring*
 - (iii) *The effectiveness of process and product changes including those arising from corrective action and preventive actions*
 - (3) *Any follow-up actions from previous management reviews*
- (b) *The management review system should identify appropriate actions, such as:*
- (1) *Improvements to manufacturing processes and products*
 - (2) *Provision, training, and/or realignment of resources*
 - (3) *Capture and dissemination of knowledge.*

Management of an organization should be clear regarding the management processes for conducting product reviews. For example,

- Are annual product review and management review of process performance and product quality the same or separate processes and if separate, how are they related?
- Which disciplines are involved?
- What is the frequency?
- What is the level of detail to be covered, e.g., when reviewing process performance and product quality monitoring data what type of data could be reviewed (e.g., sampling plans, quality attribute acceptance criteria, process parameter values, model performance) and by which group (e.g., manufacturing, quality control, local to site, across sites).

Defining the role of a Quality Steward may be beneficial.

Product quality issues from internal and external sources to be considered in the management review process may include:

- OOS and OOT events:
 - OOS and OOT quality data should be discussed and linked to process performance data
- Deviations or non-conformances and related investigations:
 - Deviations should be reviewed holistically and in comparison to other lagging quality data inputs such as complaints. Particular attention should be paid to recurring deviations as an indication of CAPA effectiveness.
- Stability results and trends:
 - An indicator of the control and consistency of any process is the product stability data. Any stability trends, deviations, or shifts since the last review should be discussed as an indicator of hidden process or product issues.

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- Product complaints and returns:
 - Details of product complaints and returns should be provided including trends. These are a lagging indicator of product quality issues, and a valuable feedback tool to be correlated with other data such as non-conformances.

Annual Product Review

The concept of regular review of drug substance and drug product quality information is described in the 21 CFR Part 211 [28] and EU [29] regulations, as well as ICH Q7 [30], e.g., ICH Q7, Section 2.5, Product Quality Review. Annual review is considered a minimum requirement.

The process performance and product quality monitoring system should be a key source of information to support the APR. Other sources include other elements of the quality system such as CAPA, change management. The major conclusions from the on-going process performance and product quality monitoring activities should be summarized as part of the annual product review. In addition, the control strategy and monitoring program should be verified, and if necessary, modified, based on the monitoring results. The conclusions of the APR also should be fed back into the PP&PQMS, where appropriate.

Major conclusions should be clearly stated in APRs regarding how selected CPPs and CQAs performed during the review period, based on monitoring results, against established acceptance criteria. Organizations may include business key product indicators in their APR process. Key product indicators are usually indicators related to business performance (e.g., yield, times of process). Reduced frequency of monitoring of some parameters or attributes could be considered for implementation for stable and capable processes. However, when results indicate that the process performance and product quality are not in a state of control, investigations could be captured in the CAPA system.

The outcome of such comprehensive review, analysis, and document routine can ensure that a lifecycle management of monitoring programs, from strategy to reporting, is performed on an annual basis as a minimum. It also can ensure that the product and process knowledge gained during the review period is captured and effectively utilized in the decision-making process for continual improvement purposes. As APR may be tied into management review activities, this approach can ensure that senior management are updated, on a regular basis, on the overall performance of process and product quality against regulatory commitments, for marketed products including issues and decisions made as a result of the monitoring programs.

3.6.2 More Frequent Management Review in Addition to Annual Product Review

There are differences between the product quality reviews described in the annual product quality review given in GMP regulations and those in ICH Q10 [1]. There is a GMP requirement that there are written documents in APRs, to be circulated for review and archived, which could lead to corrective actions or revisions to control strategy elements. However, for many organizations, an annual review does not provide real-time continual improvement opportunities. The management reviews described in ICH Q10 [1] may be more frequent and are usually conducted as meetings to share information, discuss issues, and their potential impact and resolution. More frequent meetings may be beneficial by allowing organizations to:

- Transform data into product knowledge
- Identify areas for improvement
- Ensure that Senior Management are engaged in allocating resources to make improvements

An APR may not provide sufficient real-time benefit for this type of action.

Several elements described as necessary for the APR by the US and EU regulations are appropriate for more frequent periodic review by management as described in ICH Q10 [1].

As ICH Q10 [1] describes having reviews at various levels of management depending on the size and complexity of an organization (number of plants at the site, drug substance and drug product, number of lots manufactured per month, etc.), the frequency and level of detail of product quality information discussed can also vary.

The highest level of management review should include, as a minimum:

- Observations and commitments from regulatory inspections and audits
- Product quality complaints and recall trends
- Conclusions of process performance and product quality monitoring
- CAPA effectiveness

The information can be presented and tracked as metrics, with particular focus on problem areas or adverse trends. The frequency of this type of meeting depends on the amount and criticality of the data to review. Meeting should be frequent enough so that the entire the process and product quality receive reasonable attention over time. Senior Management should be aware of and be able to react to any issues that need to be addressed, and assure that appropriate resources are assigned.

The more frequent meetings should include detailed information for critical analysis and the potential for raising opportunities for process or product improvements. Information should be sufficiently detailed to provide for effective discussion and lead to meaningful continual improvement activities, as well as addressing production issues in a timely manner to avoid repeat issues and impact to subsequent batches. The level of participation in these meetings should have sufficient technical contribution to identify issues and determine potential corrective actions.

3.6.3 Monitoring for Multiple Manufacturing/Packaging/Testing Sites

Due to business continuity factors, capacity constraints, or business strategy decisions, the same pharmaceutical drug substance and/or drug product can be manufactured, packed, or tested in more than one location around the globe. This may be done under a single set of quality specifications or different specifications as required by local registrations or regulations.

It is recognized that differences in facilities, equipment, and instrument design and construction, environmental factors and controls, raw material and component supplier sites, personnel and operating procedures exist from site to site and may impact process and product consistency and performance. Any variability arising from these differences should be understood, and where it exists, account for any impact on product quality. For each site, a control strategy for a given product should be developed based on achieving product with the desired CQAs. Based on this work and shared product and process understanding, each site should establish its own procedures for process performance and product quality monitoring (for the unit operations that the site is accountable for). It may be that the control strategy (and design space, if proposed) is the same across manufacturing sites, but this is not always the case.

When a product is manufactured, tested, or packaged on multiple sites a holistic view across all manufacturing sites should be taken, particularly when evaluating changes either at one site or across multiple sites. A change required at one site may be required at another site as well. Other events that occur (deviations, batch failures, etc.) should be evaluated across the sites for applicability of any CAPAs or process improvements both within the site where the event occurred and also across sites.

When the same unit operations, packing operations, or testing procedures are performed at more than one location for the same product or product family, quality review procedures should be established that allow for a broad review of overall process and product performance.

Periodic reviews should be performed and documented for CQAs. The objective of a periodic review includes the identification of process or product inconsistencies, trends, or other factors that may demonstrate that the process or product under review differs significantly from site to site. These differences can result in proposals for:

- Process control limit changes
- Engineering control investments
- Other improvements

The results of periodic cross site review should be evaluated and assessed relative to the need to develop corrective action plans to ensure site-to-site process and product consistency are achieved. In those cases where it is not possible to achieve consistency in process or product performance factors, the technical, engineering, and/or quality rationale should be documented and reviewed by management.

There should be a clear rationale to explain why differences in product and process performance at a different site are acceptable. A mechanism for sharing process and product continual improvement initiatives also should be established for sites sharing common unit operations or activities.

3.7 Provide Knowledge to Enhance Process Understanding

Information related to products, manufacturing processes and components should be systematically acquired, analyzed, stored, and disseminated as part of knowledge management. Sources of such information include:

- Development studies
- APR
- Product Quality Review
- Output from a PP&PQMS
- Prior knowledge of similar products and process
- Literature
- Regulatory interactions
- Knowledge from troubleshooting exercises and continual improvement activities
- Current output of risk management exercise on product and process, and on the control strategy

This information should fall within the organization's knowledge management system which should enable review of commercial manufacturing data and re-evaluation of the risk assessments for product/process. The review should allow optimization of the attribute or parameter criticality classification and continual improvement of the control strategy. Knowledge should be managed:

- Within a single product across its lifecycle
- Across products that are similar and operate using the same technology or formulation platform, and across multiple sites

Knowledge management should be facilitated using management processes such as use of quality steward roles, product quality councils, and senior management reviews. IT solutions also may play a role in an organization's knowledge management approach. In addition to availability of this knowledge within commercial manufacturing, this knowledge should be fed back into the pharmaceutical development/technology transfer groups. This can serve as prior knowledge and can help to assess risks and facilitate better classification of CQAs and CPPs in future products. For further information, see Section 4 of the Guide.

Forms of summarizing a large amount of product and process knowledge are:

- Process and product model, which may or may not be used as part of a design space
- Output of risk management exercises

It should be noted that the effectiveness of these approaches to knowledge management rely on maintaining the model or risk management exercise as current.

3.7.1 Model Effectiveness and Improvement (Enrich Design Space)

Design spaces can be based on scientific first principles and/or empirical models, hence “enriching design space” could be considered as model improvement. It is not necessary to propose a design space in a regulatory application, even though a process model may be developed.

For further information on development of models, see Part 1 of this Guide series – Product Realization using Quality by Design: Concepts and Principles [5].

In the ICH Points to Consider document [31], there are sections providing guidance on:

- Categorization of models
- Developing and implementing models
- Model validation and model verification during the lifecycle
- Documentation of model-related information

These sections are particularly helpful for implementation, validation, and verification of models.

Once a model is implemented as part of a control strategy, its performance should be monitored to ensure that it works as desired, and is improved as necessary. In a control strategy, if a model is used it is likely (in ICH terms [31]) to be a:

- Type II medium-impact model:
 - Such models can be useful in assuring quality of the product, but are not the sole indicators of product quality (e.g., used for most design space models, many in-process controls).
- Type III high-impact model:
 - A model can be considered high-impact if prediction from the model is a significant indicator of quality of the product (e.g., a chemometric model for product assay, a surrogate model for dissolution).

Models include those used to assist with process prediction and also those used in PAT analytical procedures, e.g., NIR.

Where a product or process model is not included in the control strategy, but is used as part of knowledge management, there is also significant value in maintaining the model during the commercial manufacturing phase of the lifecycle.

3.7.1.1 *Model Effectiveness Monitoring*

Results predicted by a model are valid only if the data fed into the model are also valid. A multivariate model will always provide a prediction value as long as the format and dimension of a new sample to predict is similar to the one used in the model. This does not guarantee that the predicted value is acceptable, and at least one of several diagnostic tools should be implemented to provide assurance that the new predicted sample is typical of the calibration set used.

Typically used methods are residual-space diagnostics that measure the residual part of the new prediction sample, and scores-space diagnostics that measure the distance of the modeled part of the predicted sample to the origin of the Multivariate Data Analysis (MVDA) model.

The practical approach is to:

- Build and validate a robust model at the beginning
- Control the variation
- Issue alerts if an unusual result is generated

Building a robust model depends on the type of model and the extensiveness of validation and challenge to the model during its development using both internal standards and comparison with external reference methods, where available. The multivariate methods commonly used to build predictive models for Real Time Release Testing (RTRT) can provide some diagnostics statistics that will give an indication on the validity of the prediction obtained (i.e., the diagnostics statistics will flag unexpected variability, outside the model validated ranges).

The effectiveness of a model should be gauged by any evidence that contradicts the model, e.g.:

- New data that is not predicted by the model
- The model's ability to predict the data used to create the model
- The false alarm rate when using the model

Diagnostics should be used to ensure a model is used within its model space. An out-of-model-space observation means that the data values fall outside the range defined in the training or calibration datasets.

If a variable is known to have impact on the model, but it is not practical to include in the model, an engineering solution to control the fluctuation of that variable should be used. If the controlled variable moves out of control, an alert should be issued and the model result should not be used.

The monitoring should be evaluated regularly and as needed. Evaluation based on risk management principles from use of the model should be used to determine if the model should be improved.

Approaches for model verification should be documented according to the Pharmaceutical Quality System (PQS) of the organization [31].

3.7.1.2 Model Improvement

The significance of deviations in multivariate metrics is dependent on the variation of the calibration data that was used to create the model. For example, if a model is calibrated from data with only common cause variation, such as desirable historical operation that produced acceptable quality product, model deviations indicate the process is operating in a different way than in the past. The deviations might still be due to common cause; however, in this case, the model will need updating.

A model may need to be improved if new variables are identified or even in the case when better performance is observed, for example, variability is less than the calibration data.

Common new variables may arise from a process change, e.g., data collected from a new sensor. The data used to create the model may not consider new variables from the new sensor. It is desirable that the different conditions are included in the original design so the model has these factors built-in. In practice, it is common to improve the model mathematically with limited amount of new data. Necessary testing should be conducted on the improved model and risk management principles should be used to decide whether the revised model is suitable for the intended use.

A better performance model (i.e., with smaller uncertainty) can be achieved by limiting the variation of process with necessary engineering process control of some variables (as in model development discussed above), so the predicted data are of higher quality (high signal to noise ratio) since the model does not have to compromise as many variables.

Model improvement itself is a reflection of process knowledge accumulation, and the knowledge could be used to improve the model further, e.g., build a more robust model or propose a new mechanistic model based on increased process understanding.

3.7.2 Regular Re-evaluation of the Risk Assessment

The realization that the risk management process itself may not be perfect is considered fundamental to a risk assessment exercise. A team may fail to identify all hazards although it is expected that the vital hazards which account for the majority of risk are identified. Estimates of probability of failure may not be accurate due to the body of data from which to draw being relatively small compared to those available years later. The real benefit from a risk assessment exercise comes from the consideration of hazards, understanding them to the highest level at a particular point in time, and prioritizing efforts to understand, control, and prevent them.

The risk assessment exercise should be periodically revisited. This can be an opportunity to train new staff and approach earlier problems with added experimental data. There also may be cost benefits. Investments made in engineering/operational process controls and knowledge gained over time can be captured during a re-assessment.

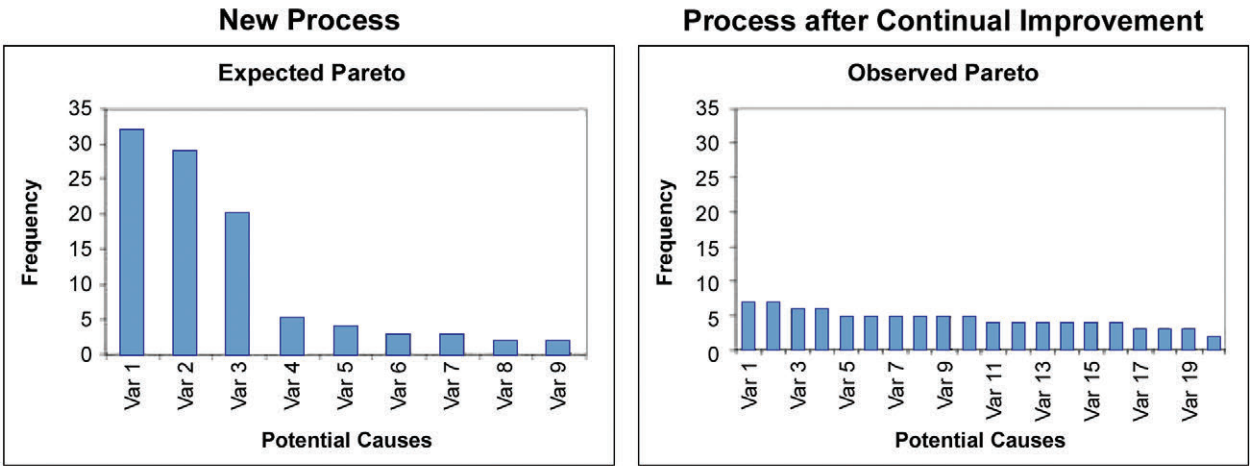
Examples of Different Approaches to Use and Presentation of Risk Management Exercises

Example 1:

Pareto charts are used to summarize output from risk management exercises. Figure 3.9 shows two plots of Risk Priority Numbers for probability of not achieving process control limits. On the left is a Pareto Distribution which is characteristic of new processes. A vital few factors (each with > 20% of the total risk, Variables 1 to 3) result in the majority of the Risk (at 80%). Such a plot is helpful in communicating what is most important and what is less important and serves as the basis for prioritization of resources for controlling the risk and improving the process and/or control strategy.

On the right is a picture of how such systems can change over time with continual improvement. As high priority risks are identified and the probability of their occurrence via process changes is decreased then the Risk Priority Number also decreases. This allows an organization to recapture an investment in engineering/operational controls by updating their control strategy and reallocating resources to the next most important risk.

Figure 3.9: Frequency of Probability of Failure to Achieve Process Control Limits for a Range of Variables



Used with permission from J. Orloff & Associates, Inc. [32]

Example 2:

Process capability measurements are used to identify the level of risk and need to improve a process. Figure 3.10 shows a process with a high probability of producing OOS with potential risk to the patient. Investment in engineering controls that lead to process improvements could result in a process with Total Variability, shown in Figure 3.11. Engineering controls are typically physical elements such as capital improvements to equipment or design features of equipment. For example, if flow rate through a pipe were a CPP, sizing the pipe and pump in such a way that it is impossible to operate out of the acceptable process range is an engineering control.

Figure 3.10: A Process with Total Variability equal to Difference between USL and LSL (Cp = 1)

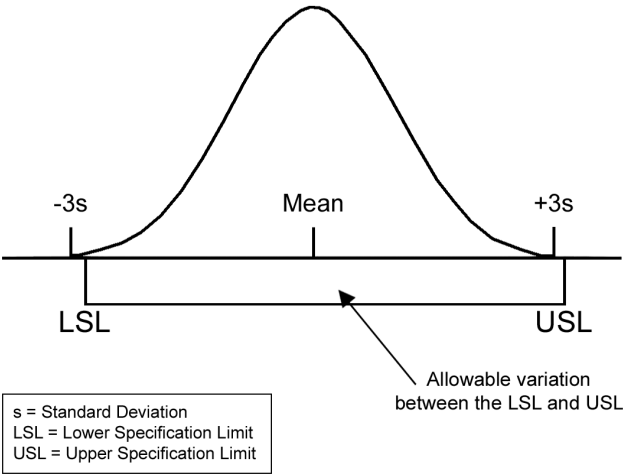
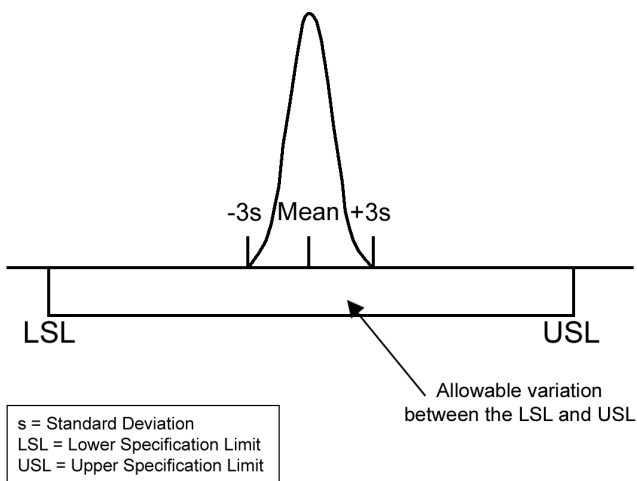


Figure 3.11: A Process after Continual Improvement with reduced Total Variability ($C_p > 1$)



Variability such as that seen in Figure 3.10 gives high probability of not meeting specification acceptance criteria (OOS). In contrast, the process with reduced variability, as given in Figure 3.11 corresponds to a decrease in the probability of an OOS and justifies a decrease in the Risk Priority Number (i.e., Risk Priority Number = Severity $\times$ Probability $\times$ Detectability) assigned to the probability of failure to comply with specification acceptance criteria. By decreasing the variation, the probability of producing OOS materials goes down.

This large decrease in risk could justify cost savings in several areas, e.g., from preventing rejected material to a re-evaluation of the control strategy. An investment in engineering process controls to improve a process may justify changes to the control strategy such that recurring costs (e.g., sampling) could be decreased and resources reallocated to reducing the probability of failure of the next highest risk.

Continual improvement can come from many different sources, but a cross-functional team effort should be employed to help to ensure success. For the team's efforts to be fully cooperative, effective communication between and within sites, and all levels of personnel should be involved to share issues, conclusions, etc. The proper communication can come from having electronic systems where all relevant sites have access to one another's documents. With data collection from process monitoring and the proper documentation, making connections between similar products and manufacturing platforms on a regular basis can provide constant and perhaps consistent improvement opportunities. In addition to data collection directly from the product or process, data collection can originate from:

- Product complaints
- Adverse event reporting
- CAPAs
- Other quality assurance documentation

All aspects of a process can be improved upon or updated. Data collection from process monitoring can be used to update models, add historical background to control charts, or be utilized in preventing or aiding in corrective actions for manufacturing or quality issues.

3.7.3 *Modern Approaches to Process Validation*

As there are advances in approaches to process validation, e.g., more use of the lifecycle approach, there is likely to be an impact on how PP&PQMSs are established and implemented.

For further information on approaches to process validation, see the ISPE Website [33].

3.7.4 *Innovation*

Innovation can be a major contribution to increasing process and product understanding and can take many forms. Thinking of the impact of innovation in terms of the PP&PQMS, there are opportunities such as:

- Use of new analytical tools
- Use of new mathematical or statistical tools or approaches
- Use of new equipment within an existing process
- Use of new engineering controls

Innovation can come from many sources, but does not necessarily mean finding something entirely new, such as complete redesign of a process; it also can come from updating processes and procedures, through continual improvement. Innovation is likely to increase product and process understanding when implemented as part of a PP&PQMS through continual improvement.

Implementation of innovative technologies may require significant product and process understanding to be successful. As process understanding increases through the commercial manufacturing phase of the lifecycle, opportunities for innovation as part of continual improvement may become apparent.

4 Roles and Responsibilities of Quality Stewards

4.1 Introduction

A PP&PQMS is a key component of product quality life cycle management within a PQS. It integrates initial process design with necessary changes in response to monitoring outcomes and need for continual improvement.

An extensive amount of knowledge and expertise in all areas of manufacturing and quality is required throughout the long term and iterative process of pharmaceutical development, technology transfer, and commercial manufacturing. At a given point in the product lifecycle, different technical and business functions are more involved than others in order to achieve project milestones and solve problems specific to the lifecycle stage of the product. Therefore, it can be particularly challenging for an organization to retain and document all the key information pertaining to the process and the product from its initial design phase to product discontinuation. Performance issues can be indicated by:

- Monitoring results or product complaints
- Investigations and remediation process
- Major changes and presumed impact
- Regulatory strategy
- Supply chain considerations

These performance issues form the knowledge history that extends through a long period of time, and possible organizational and ownership changes. Large organizations with a rich pipeline, where multiple products may be at the same stages of the development and lifecycle, are particularly challenged by their ability to manage the knowledge base effectively, in order to maintain continual oversight and consistent decision-making. The concept of the role of Quality Steward came about to assist organizations manage this complex process.⁵

The concept is equally applicable to:

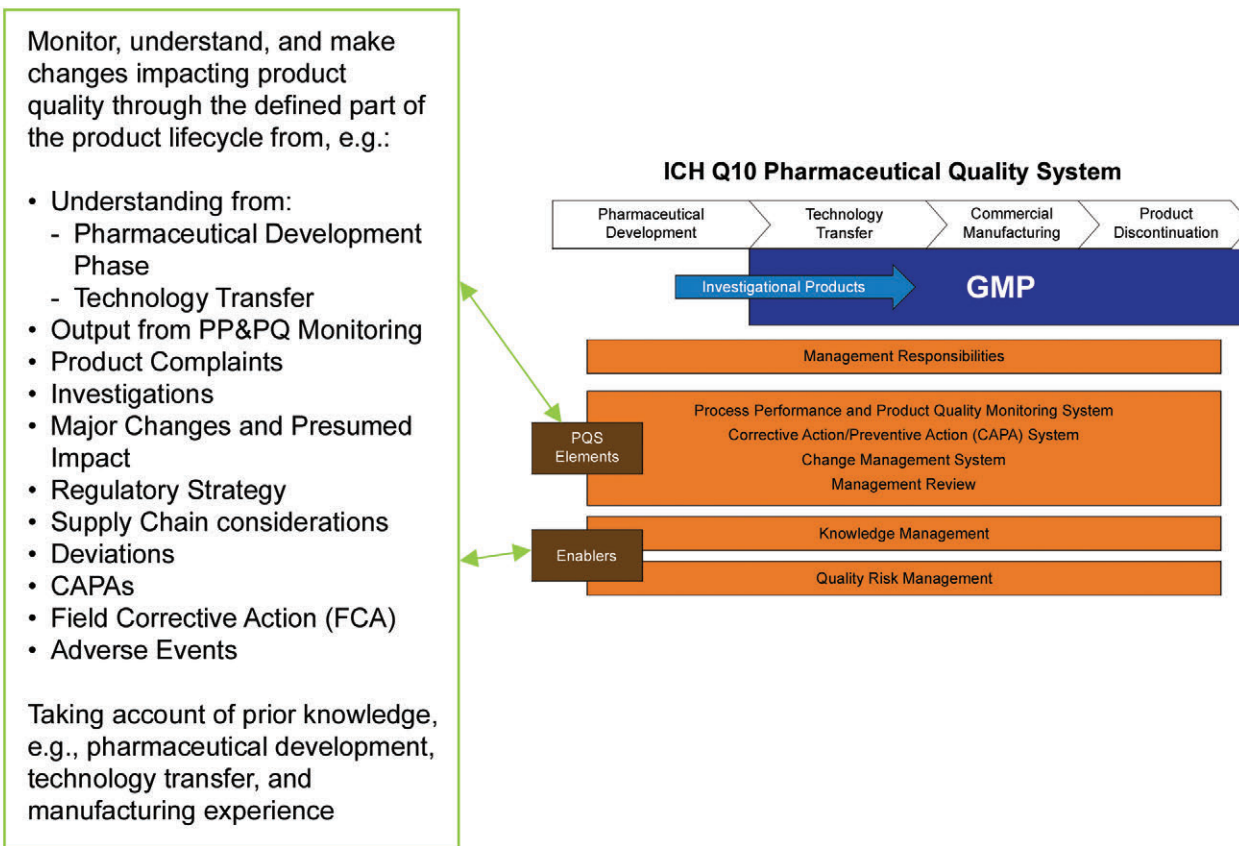
- Small organizations with a relatively small pool of knowledge
- Generic organizations
- Innovator organizations with large portfolios of existing, marketed products
- Organizations with a diverse supply chain extending across many suppliers, sites, and countries

4.2 Quality Steward Role

A summary of the principles of Quality Stewardship is given in Figure 4.1, which shows at the top of the box on the left the high level role. The bullets show examples of elements impacting product quality with the arrows linking to the diagram of the ICH Q10 PQS model [1].

⁵ Note: there is no current requirement in regional GMPs or international guidance for the provision of a Quality Steward (at time of publication).

Figure 4.1: Principle of Quality Stewardship



The role of a Quality Steward can involve accountabilities and responsibilities being delegated in different ways across different disciplines in different organizations. Names for a key role with an oversight of responsibility for one or a series of products could include:

- Product Quality Stewards (and sub-categories, such as Change Process Steward, Clinical Quality Steward, and Commercial Quality Steward)
- Product Design Owners
- Product Quality Leaders

Although varying in scope of responsibilities, organizational affiliation and titles from organizations to organizations, all these groups are composed of quality professionals who have the oversight and overall decision-making roles for product quality. The individuals in these functions may be responsible for one or several products. In some organizations, two different Quality Steward Organizations exist:

1. One for development products
2. One for marketed products

In other organizations, only one function is responsible for overseeing product quality over the entire product lifecycle starting from clinical development stage.

The role of Change Process Steward is a focal point for:

- The management of change
- The portfolio of change proposals and plans
- The measurement of the effectiveness of change

For further information on the role of Change Process Steward, see Part 3 of this Guide series – Product Realization using Quality by Design: Change Management System [9].

The role of Change Process Steward may be combined into the role of a Quality Steward.

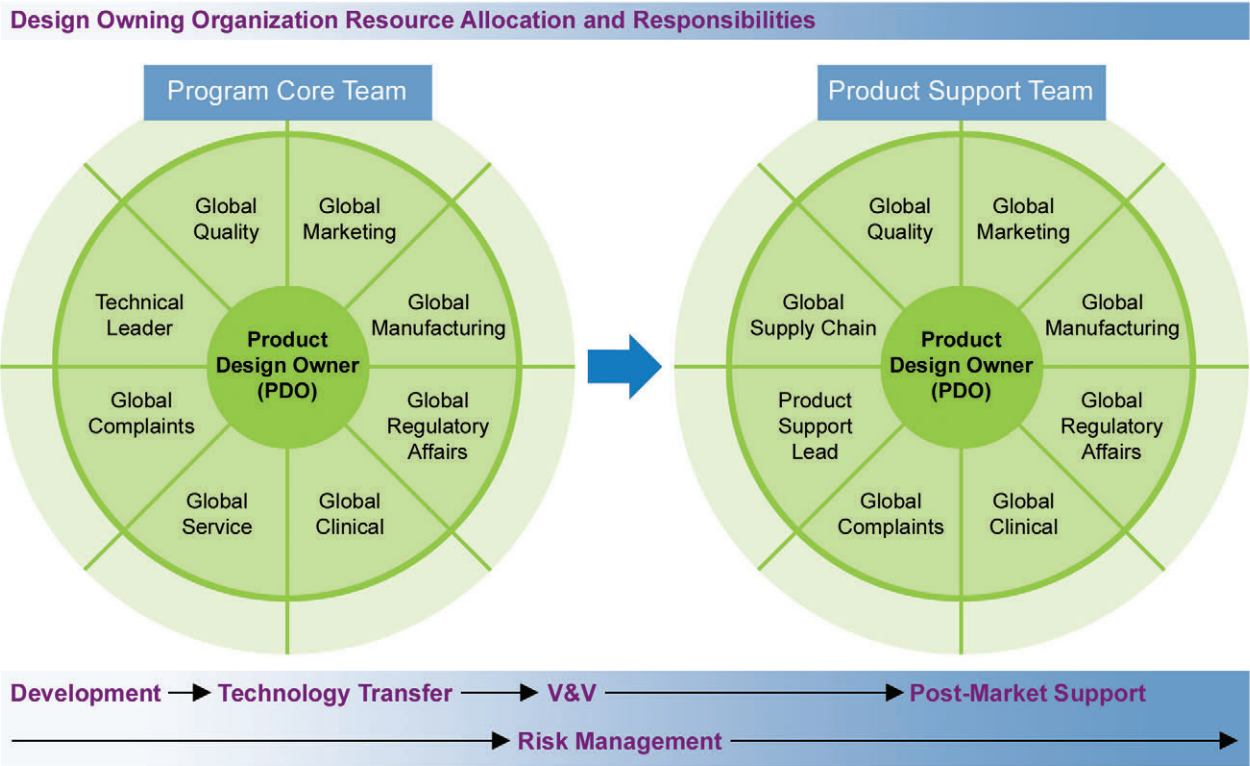
4.2.1 Example Quality Steward Role 1 (Product Design Owners)

In this example, Product Design Owners (PDOs) have global, and ultimate, responsibility and decision-making authority to monitor and make changes throughout a product lifecycle (Development through obsolescence) commensurate with the risk of quality indicator monitoring. A PDO cannot make all decisions by themselves, but rather, is supported by a cross-functional team that reviews data to identify any signals that need to be addressed. The PDO relies on cross-functional involvement during development through post-launch, which includes establishing and monitoring quality indicator data sources, analyzing/trending of quality data, and taking actions when appropriate. The quality data monitored includes:

- Deviations
- CAPAs
- Manufacturing/quality control result trends
- Field Corrective Action (FCA)
- Complaints/adverse events
- Any information relevant to the Risk Management File, such as hazards, risk controls, and tools used to identify these items

Other sources of process performance and product quality data that are part of the PDO's toolkit for a comprehensive product review also exist. Management review and annual product review are also pertinent data sources that influence a PDO's decision-making. The interaction of PDO function with their key stakeholders from product development stage to post-marketing product support is shown in Figure 4.2.

Figure 4.2: Quality Steward 1, Roles, Responsibilities, and Interactions



4.2.2 Example Quality Steward Role 2 (Product Quality Steward)

In this example, two Quality Steward (QS) groups can exist within the global organization:

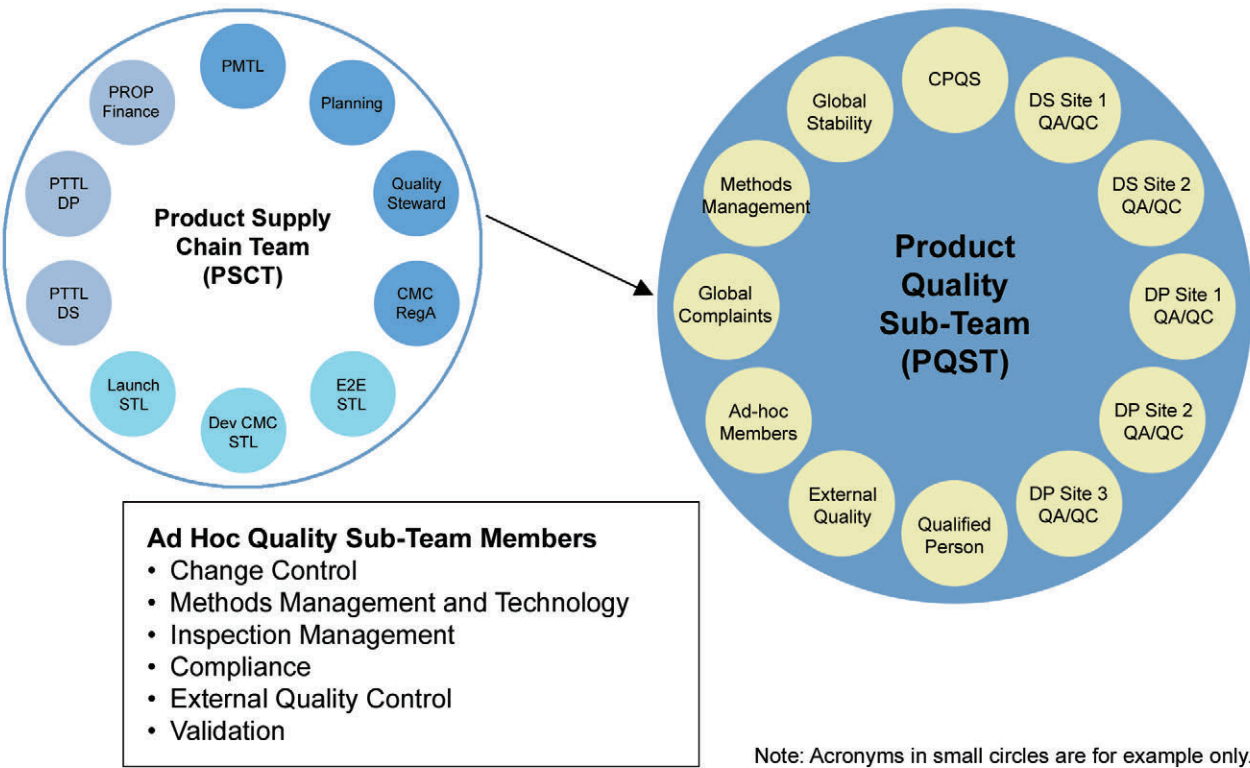
1. Clinical QS group is part of the development organization
2. Commercial QS is part of the commercial product quality management organization

Clinical QSs are responsible for coordinating and overseeing all quality activities and decisions from preparation of toxicology study material to commercial product launch. During the course of launch preparation, they transition their responsibilities to the commercial QS organization. Clinical QS function represents all areas of quality, including various quality unit and quality system functions, on Chemistry and Manufacturing and Control (CMC) teams. They are dedicated to the management of quality activities during the manufacturing process development. They serve as the quality single point of contact. They communicate product development timeline and issues to quality to ensure successful manufacturing, product testing, and release without compromising product quality, and regulatory submission. They participate in regular technical team meetings and provide clear and timely feedback to technical teams on behalf of quality. They work with groups both within and outside of the quality organization of the company, including external partners and Qualified Persons. They are responsible for establishing quality documents in the quality system (e.g., product specification, GMP stability protocol), required per company Quality Policies and Procedures.

Commercial QS acts as the voice of quality (quality assurance, quality control, and quality systems) and provides quality expertise and leadership on the commercial product supply chain team (PSCT, see Figure 4.3). They serve as the communication conduit between PSCT and quality functional areas/departments. They ensure quality requirements are met during PSCT-driven activities and changes. Additionally, they drive changes to ensure appropriate product quality lifecycle management spanning across the network, contract sites, and partners.

Specifically, they develop and manage the implementation of quality-related activities such as end-to-end risk assessment and resolution of identified risks. They are responsible for overall product quality monitoring and reporting. They monitor and report on product quality issues and overall “product health” status, across the entire commercial manufacturing and testing network, on a monthly basis. They partner with quality subject matter experts, such as those for stability, raw material, product complaints, and quality control analytical method, to resolve product quality issues through investigations and change management. They participate in Annual Product Quality Review as one of the overall reviewers and approvers.

Figure 4.3: Product Quality Sub-team and Relationship of Quality Steward with Commercial Product Supply Chain Teams (PSCTs)



Appendix 1

Considerations for Analytical Procedures and Data

5 Appendix 1 – Considerations for Analytical Procedures and Data

According to ICH Q6A [10] and Q6B [11], it is a requirement to have a finished product specification [34] established and complied with, when tested. Some analytical procedures in finished product specifications are applied during stability testing [31, 34]. It is also likely that analytical procedures in specifications will be the reference procedures when a “secondary” procedure such as NIR spectroscopy is applied.

Measurement of parameters and attributes is important to establishment of a PP&PQMS and the analytical procedures and associated data collection and manipulation systems, if used, should be carefully assessed to understand the quality of data being generated. There are several factors that should be considered in how analytical procedures are operated, data collected and manipulated (e.g., traditional laboratory testing with manual data collection and calculation of results vs. real time release testing using process analyzers, process controllers and computerized data management and storage systems), e.g.:

- Measurement near or in the process
- What end-product testing is required?
- Are in-process samples needed?

The control strategy and its application to the overall process is the basis for defining this.

Measurement of attributes and parameters in a PP&PQMS can be achieved by the following procedures, the first three being considered PAT procedures:

- On-line measurement: this is automated sampling where the sample is diverted from the manufacturing process, and may be returned to the process stream, i.e., direct transfer of material is made to the instrument from the process. An example of this would be where a diverter valve could siphon off an in-line sample directly to an installed interface to provide a measurement of the analyte without disrupting operations. Also, this could be simply an apparatus directly placed into a vessel or chamber to directly measure the process.
- In-line measurement is defined when the sample is not removed from the process stream and can be invasive or noninvasive, i.e., when the sample interface is directly placed into the vessel for continuous measurement.
- At-line measurement occurs when the sample is removed, isolated from, and analyzed at close proximity to the process stream (unit operation being performed).
- Off-line measurement is when the sample is collected, separated from the process, and analyzed away from the process stream, e.g., in a quality control laboratory remote from the unit operation.

Note 1: Gas Chromatography (GC) and High Performance Liquid Chromatography (HPLC) are typical off-line methods that are performed in a laboratory setting. Newer technological advances have enabled these technologies to become more portable and robust to allow at-line or perhaps even on-line use.

Note 2: how the sample is taken does not reflect the time taken for the analysis to be completed and results become available. Timing of availability of results depends on the speed of the analytical procedure employed.

Several types of process analyzers exist for conducting on-line, in-line, at-line, or off-line testing of samples. There are distinct differences, advantages, and disadvantages of each. Discussion of all possible procedures for analysis is beyond the scope of this Guide. Some examples of potential analyzers with their limitations are given in Appendix 3. Selection of the most appropriate analyzer technology can be based on many variables, but some of the more commonly derived criteria are:

- Phase of the analyte (e.g., liquid phase, semi-solid, or solid state).
- Type of analyte and the limit of sensitivity/detection needed to determine concentration.
- Capabilities of analyzers to be used in GMP environment
- Environmental considerations such as the amount of background light
- Refractive and reflective properties of the vessel that contains the material, and indigenous to the actual material

There are several general factors that require consideration which impact on attribute measurement:

- a. Based on the type of analytical procedure applied and associated measurement system and technology being employed, validation of the procedure is expected [10]. Depending on the analytical procedure, validation characteristics could include:
 1. Precision, e.g., repeatability over a short period of time (e.g., noise) and precision over a more extended period of time (e.g., drift), and between days, analysts, equipment (intermediate precision) and between laboratories (reproducibility).
 2. Accuracy
 3. Specificity
 4. Limit of detection and limit of quantitation
 5. Linearity and range
 6. Robustness of the procedure which for on-line and in-line procedures (see later) often depends on equipment performance

Thus, part of the PP&PQMS establishment plan should be validation and/or re-calibration of components, processors, controllers, and the like to assure the quality of data being generated.
- b. Care also should be taken when reporting data that an appropriate number of significant figures are being used based on the precision of the method.
- c. Environmental considerations. When using highly sensitive equipment environmental conditions should be taken into consideration. For example, if Programmable Logic Controllers (PLCs) are being utilized, humidity and temperatures should be considered important and controlled as needed. Choice of controllers, e.g., in terms of robustness of instrument should be carefully considered.
- d. Location of measurement device. A determination should be made prior to execution as to where the monitoring devices are located. The question should be asked “does the measurement instrument analyze a representative sample?” Location also is statistically dependent on measurement response time and signal-to-noise ratios.

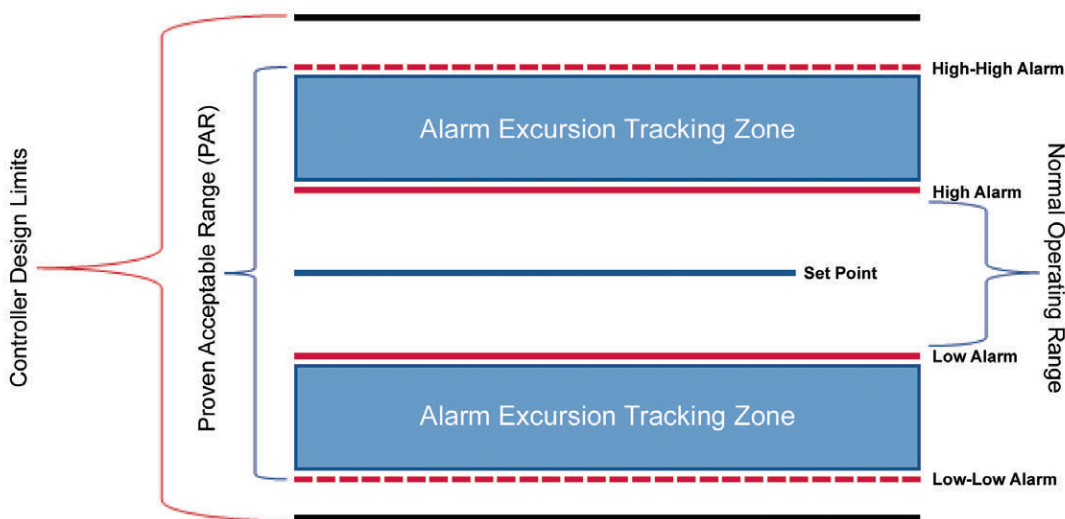
- e. Frequency of sampling. This depends on whether samples are being taken for off-line analysis, or whether an at-line or in-line procedure is being used. Sample size (if possible) and frequency of data collection should be defined. Frequency of data collection should be sufficiently fast to detect frequency of change of a signal and if the data collection rate is insufficient the detected data may not represent the signal, a phenomenon known as “aliasing.” Therefore, issues to consider are rate of production of data so that a minimum frequency of data collection can be selected where aliasing can be avoided.
- f. Cost. This is always a factor and justification of cost is a natural challenge to good project managers leading these types of efforts. However, consideration should be given to value, for example long term sustainability (total cost of ownership) instead purchase of a less robust, cheaper option, which may lead to more repair costs and operation downtime. This is after all, an investment in the overall product quality and assurance. A cost benefit analysis should be performed to assess required assets needed for implementation and operation. Consideration also should be given to capacity and modifications for the longer term. Typically, the more customizable a monitoring device is, the higher the total cost.
- g. Speed of response from the monitoring device. It is beyond the scope of this document to take an in-depth look in to what the industry refers to as “signal-to-noise response,” but for general understanding an awareness should be made that a monitoring device has a certain speed for detecting a signal from a process analyzer. It takes time to send a signal from the process analyzer to a monitoring device whereby an interface provides a graphical representation of the numerical value to the end user. How quickly a measurement must be made and recorded are key criteria as to the type of monitoring device that should be chosen.
- h. Rate of change. The rate at which a chemical, mixture or substance changes will potentially impact measurement readings. Factors must be considered when measuring in situ such as the effect the physical manufacturing process has on the analyte to be detected, and the sensitivity of the monitoring device to pick up on changes.

Further suggested considerations include:

- Location and type of sensors, controllers and like apparatus should be considered early in the development stages of the process. This can make future process optimization more cost efficient and less complex. For new products, sensors and controllers should be implemented as early in the development process as possible.
- Procedures should be considered for process and automation maintenance, calibration, and overall operation. Training and visual aids typically assist in this process. Operator skills should be considered early on in the development process.
- Operational and testing protocols should be realistic in that, e.g., requested parameters of measurements must not go outside the range of the instrument specified. This should have been identified in the design qualification/developmental/commissioning/performance verification stages of the instrument.
- Operational ranges should be shown capable of being maintained as long as would be necessary during routine production.
- Measurement errors of process analyzers and process controllers should be understood and communicated.
- Device “noise” and “lag” are again common and the limits of these must be understood as part of development and prior to fully validating the process. Issues such as these need to be understood and corrected prior to the Operational Qualification (OQ)/Performance Qualification (PQ) stages. By not proactively understanding and correcting these issues prior to commencing qualification, it could result in costly retrospective studies, and/or repeats of design qualification.

Figure 5.1 illustrates two concepts for depiction of a basic process control system to evaluate blend homogeneity of a solid dose drug product: showing limits where an alarm is triggered (Low and High) based on limit of detection of the process control system and normal operating range, and a wider set of limits (High-High and Low-Low) based on proven acceptable range.

Figure 5.1: Basic Control Design



The sensitivity of the process sensors and controllers should be set within the operating range with alert and action limits being set for low and high limits. The purpose of setting these ranges is to ensure that the anticipated measurements taking place during the process do not exceed the PAR of the specific process analyzer, i.e., the process analyzer has sufficient range to measure the parameter or attribute in question with sufficient accuracy and precision. When designing the process control and measurement system, there should be sufficient process understanding and the anticipated attribute and parameter ranges should be known prior to purchasing equipment, so that the system is designed based on known User Requirement Specification (URS) and Functional Requirement Specification (FRS).

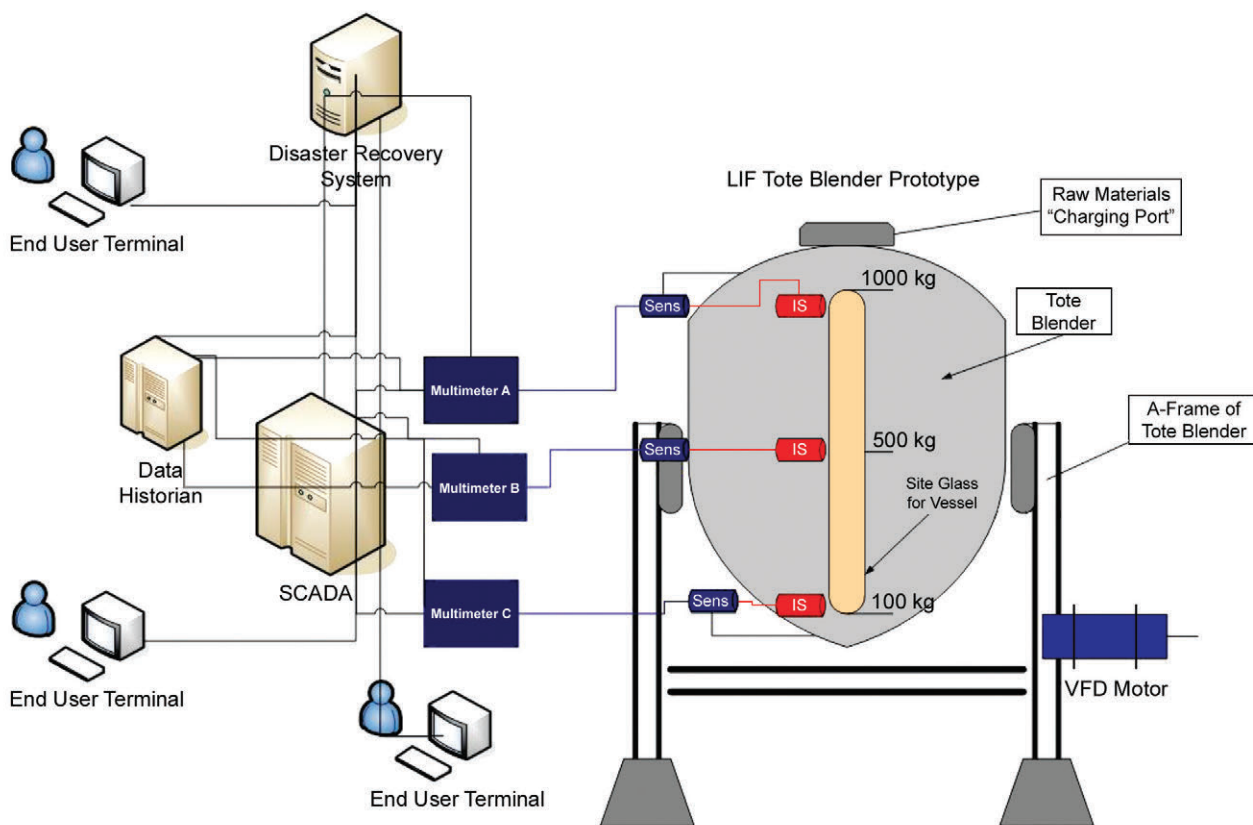
Figure 5.2 shows a complete system mock up for process control system for the blend homogeneity element of a control strategy for a granulation operation. This operation uses Laser-Induced Fluorescence (LIF) technology to determine homogeneity of the granulation blend. Some key messages from this example that should be applied to processes include:

- Sensor placement. It should be representative of an entire batch including the upper and lower extremes where homogeneity may not be as robust (e.g., air pockets can develop and granulation fines may be dislodged to the bottom of the tank).
- Process sensor feed rate into control devices such as multi-meters. These devices are required to translate the signal detected from the sensor to a graphical user interface for interpretation. Devices such as multi-meters also minimize lag and noise variables to ensure measurements are near real-time and representative.
- Process control system elements such as:
 1. Redundancy of system and sensor failure counter measures
 2. Sensor protocol interface and network robustness (similar to “sensor feed”)
 3. Hardware/software interoperability analysis
- Graphical user interfaces may be placed in various locations near the unit operation to enable an operator to see how the process is performing. Suitable alarms may operate in parallel to ensure when limits are exceeded; the operator is aware and can react accordingly. These interfaces also should enable process drift to be detected prior to it becoming a quality issue.

- Supervisory Control and Data Acquisition (SCADA) devices that may be built in (or integrated) into an existing process control system. SCADA Systems are typically centralized systems that monitor and provide process control for an entire host site of sensors and/or controllers. For example, a site that contains many Programmable Logic Controllers (PLCs) may typically be run through an industrial process control system on a daily basis. However, when there is a need to intervene or deviate from the normal path to change set points, interface or other pre-programmed conditions, SCADA allows the operator to manually intervene into the process thereby overriding the industrial controller. Systems should have a disaster recovery process in place where automation fails or control plans should be enacted.

Validation of the process control system, as an entire system, is key to ensure that the above interfaces, peripherals and sensors operate in a validated state as a closed loop system and demonstrate compliance with 21 CFR 211 [28], 21 CFR Part 11 [35], and EU GMP Volume 4 Annex 11 [36]. Valuable guidance on how to validate process control systems and relevant computer systems is given in ISPE GAMP® 5 [25] and the ISPE GAMP GPG: A Risk-Based Approach to GxP Process Control Systems [26]. Consideration in the master validation plan is required regarding when and how upgrades will occur if software is part of this system to ensure that a patch upgrade to one part of the computerized system does not affect up or downstream operations. Data recovery and integrity plans also should be established in the context of disaster recovery.

Figure 5.2: Process Control System of a Blending Operation



Robustness of Analyzer

For process analyzers where the procedure is developed as a secondary procedure utilizing a different method/measurement approach as the primary reference, it is good practice to keep reference method for testing for attributes and parameters in parallel until the control strategy is fully verified. Once verification has been completed, a risk assessment can be performed to evaluate either elimination of the reference procedure or reduced frequency of testing; however, account has to be taken of commitments made in regulatory applications relating to RTRT, where RTRT is proposed as a control strategy element, as well as cGMP requirements

For example, it is a requirement to have a finished product specification established and met when tested according to ICH Q6A [10] and Q6B [11] and to have analytical procedures available for stability testing. It is likely analytical procedures in specifications and applied to stability samples will be the reference methods, e.g., as given in pharmacopoeias and suggested.

Other important points to consider particularly relevant to PAT procedures are:

- Keeping reference laboratory procedures in place in case of a control system deviation at the shop floor can serve as a backup to the main control plan.
- It should be assured that the sampling plan for the reference procedure provides the same level of assurance as that of the PAT measurement and that appropriate procedures are followed to investigate process control system deviations, see ICH Q8, Q9, and Q10 Q&A [34]. Although end product testing cannot automatically be substituted when PAT procedures “fail” the outcome of an investigation may require some form of testing using the reference procedure to assure that a batch release decision complies with the content of the marketing authorization and GMP.
- Establishment of a system suitability test, which ensures that the analytical instrument is accurately measuring.
- Level of validation of PAT procedure. The level of validation for any given analytical procedure should depend on the attribute/functionality that is being measured. In other words, the level of validation should be commensurate with the level of risk associated with the use of the specific PAT procedure.
- Inclusion of PP&PQMS and associated control plan in the PQS, which should lead to their inclusion in the internal audit program.
- Data or summaries of data from the PP&PQMS should be included in the management review and/or product quality reviews.
- Technological upgrades needed either in the laboratory or manufacturing facilities to enable the technology to be used and its impact on the measurement.
- Assignment of appropriate security level passwords or biometric challenges to ensure data integrity remains in place.
- Disaster recovery

There are unique and distinct advantages gained by designing in measurement control systems into the process and moving toward RTRT as opposed to traditional approach based on sampling for off line testing using batch laboratory analytical procedures. Some advantages are:

- Economies of scale
- Continuous product monitoring

- Ability to react timely to process drift observed from the real-time measurements
- Decreased product cycle time for manufacturing
- Lower costs of quality, e.g., the ability to detect and correct potential failure immediately vs. much later in the process
- Lower carrying costs of inventories for manufacturing sites due to quicker turn rates of finished goods

Training and appropriate documentation should be part of the support for the deployment and application of the control strategy within the PP&PQMS, so that personnel have the appropriate tools and training to execute the control strategy to assure the desired state of control.

Appendix 2

General Considerations for Establishing a Control Strategy

6 Appendix 2 – General Considerations for Establishing a Control Strategy

General Notes and Considerations

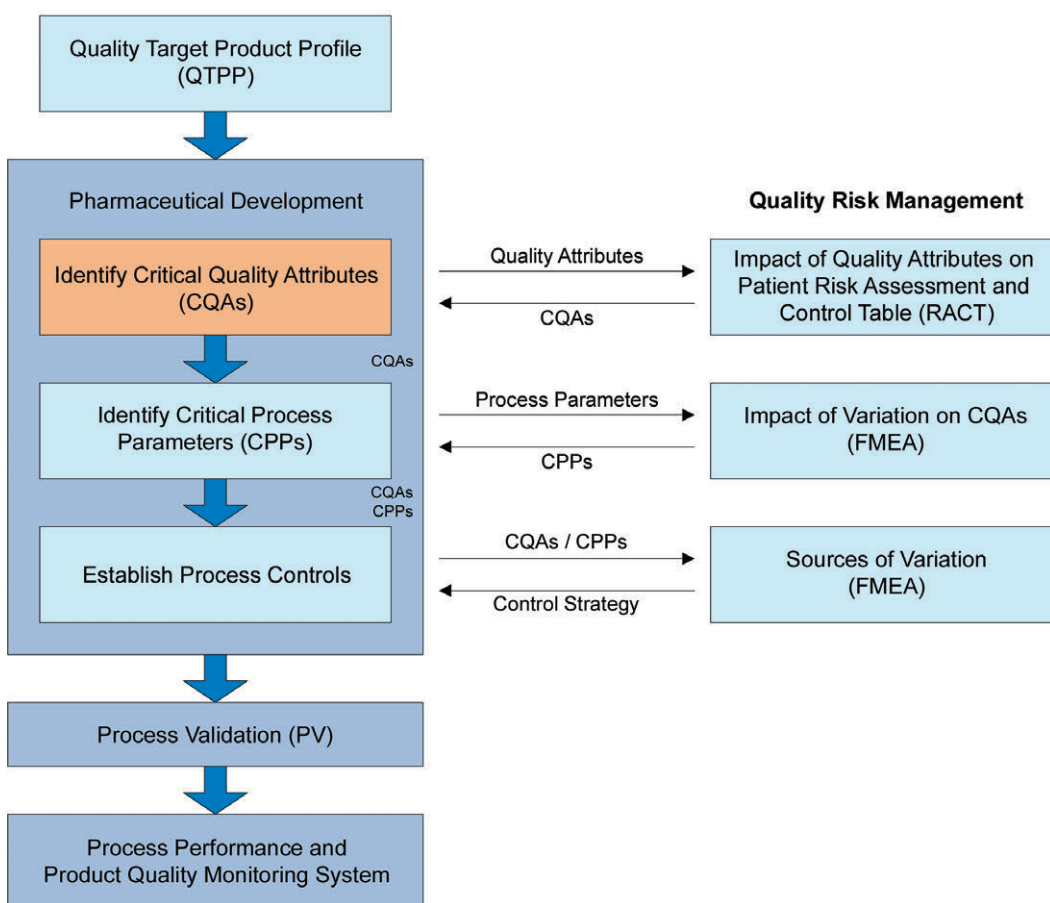
- a. Prior to the establishment of any control strategy and sustainable PP&PQMS, consideration should be given to the types of products in scope and the overall validation or development strategy for a particular product.
- b. A scientifically sound control strategy is an integral part of science- and risk-based, strategic validation planning. The control strategy can be leveraged during validation maintenance activities, such as to justify the scope of revalidation required in the event of a change or to provide data for periodic review to demonstrate that the process remains in a state of control. It provides data to support process capability measurements.
- c. When evaluating existing products, the current control strategy, whether derived from an enhanced or minimal product and process development approach, can be used as the foundation for the PP&PQMS requirements for the product. Evaluation of process performance and product quality data can facilitate better process understanding and continual improvement of the control strategy. For further information, see the ISPE paper, Application of Science- and Risk-based Approaches (ICH Q8, Q9, and Q10) to Existing Products [37] for recommendations of implementation and three case studies.
- d. The most important facet of control strategy is the proper identification of a product's CQAs and process parameters and material attributes impacting on those drug product CQAs. Process parameters can be assigned as critical and there is "continuum" of criticality based on risk from those judged high to those medium or low. For discussion on the topic of "criticality" as relevant to critical quality attributes and critical process parameters, please refer to ISPE Guide Series: PQLI® from Concept to Continual Improvement Part 1 – Product Realization using Quality by Design (QbD): Concepts and Principles, Topic II, Criticality [5], and for exemplification to Part 2, Product Realization using Quality by Design (QbD), Illustrative Example [6]. Drug Product CQAs are considered an essential basis of design for the control strategy.
- e. It is highly recommended that a control strategy is developed and updated in accord with the current state of process and product knowledge and understanding as part of continual improvement. Given that the control strategy is likely to be improved, supporting technology platforms should be assessed in parallel using risk management to determine the most appropriate infrastructure. Once any new specific attribute and/or parameter has been identified as critical then optimum user requirement specifications for supporting relevant measurement and/or process control technology can be proposed.
- f. Developing a robust process control and monitoring strategy should include several disciplines and cross-functional working in order that the process and product holistically can be reviewed so design-in/design-out engineering can be achieved. These functions can include, but are not limited to:
 - R&D scientists
 - Engineering
 - Statisticians
 - Automation and technology
 - Manufacturing
 - Quality

- Validation engineering
- Chemometricians

Part 2 – Product Realization using Quality by Design (QbD): Illustrative Example [6] provides comprehensive exemplification of development and transfer into manufacturing of a control strategy for some steps in drug substance synthetic process and a tablet manufactured using a simple blend and direct compression process. It focuses on those elements of the control strategy developed using enhanced, QbD approaches. It does indicate the complexity of a control strategy and gives examples where expected GMP procedures are important elements in a control strategy.

Parts 1 and 2 of the Product Realization using Quality by Design guides series [5, 6] provide extensive discussion of the enhanced, QbD approach. A further example of a company approach is given in Figure 6.1, concentrating on those elements developed using an enhanced, QbD approach. From these activities a comprehensive PP&PQMS, e.g., for in-process materials, process parameters, or finished drug product can be established.

Figure 6.1: A Company Enhanced, QbD Approach to Development Leading to Parameters and Attributes to Include in a PP&PQMS



These steps are often performed iteratively during the development process. An early goal in development is to appropriately assess risk for each CQA prior to establishing controls.

Once CQAs have been established, then process parameters and Material Attributes (MAs) impacting CQAs can be developed. Much of this information can be derived from laboratory and scale-up/scale-down activities as discussed particularly in Part 2, Product Realization using Quality by Design (QbD), Illustrative Example [6].

After the process and product has gone through the process design finalization phase during pharmaceutical development and prior to technology transfer an overall Process Flow Diagram (PFD) could be established. The process flow diagram helps articulate process understanding and serves as an invaluable training and communication tool throughout the lifecycle of the product. Process flow diagrams should include, but are not limited to:

- a. Engineering symbols and conventions to illustrate universal understanding or unit operations
- b. Process stage
- c. Inputs, outputs and waste work streams
- d. Equipment train and set flow charts
- e. Unit operations
- f. Controls that are needed to run a specific operation
- g. A general description of the process
- h. Depiction of flow of raw materials
- i. Environmental control factors
- J. Depiction of where product or material is tested and how tested
- k. Proven Acceptable Ranges (PARs) or design space(s), as appropriate
- l. Notations of steps to communicate if the process is fully automated, semi-automated or manual, or a combination of the above

Appendix 3

Analytical Procedures Potentially Used in a PP&PQMS

7 Appendix 3 – Analytical Procedures Potentially Used in a PP&PQMS

Table 7.1: Commonly Used Types of Analyzers, Basic Capabilities, Limitations, and Highlights

Sensor Family Group	Analyzer Technology	Nomenclature and Description
Light/Radiation Based	Near Infrared (NIR)	Specifications: - Wavelengths 4,000 to 12,500 cm^{-1} - Glass and quartz are transparent to NIR Light - Pathlength 1 mm to 1 - 5 cm Advantages: - Relatively fast - Sampling and sample prep not typically performed - Non-destructive for sample Disadvantages: - Newer technology - Complex algorithms/quantification
Light/Radiation Based	Infrared (IR)	Specifications: - Wavelengths 400 to 5,000 cm^{-1} - Optics can be expensive and made of exotic materials - Pathlength is 0.1 mm Advantages: - Relatively fast - Sampling and sample prep not typically performed - Non-destructive for sample Disadvantages: - Newer technology - Complex algorithms/quantification - Difficult to measure in water based systems Infrared (IR) technology is based on the principle of absorbance whereby as the IR light source passes through a sample, the light absorbance is compared to a known standard.

Table 7.1: Commonly Used Types of Analyzers, Basic Capabilities, Limitations, and Highlights (continued)

Sensor Family Group	Analyzer Technology	Nomenclature and Description
Light/Radiation Based	Fourier Transform Infrared (FTIR)	Specifications: - Internally calibrated against HeNe laser for longer term stability Advantages: - Relatively fast - Sampling and sample prep not typically performed - Non-destructive for sample - Mechanical simplicity - No affect from stray light in background Disadvantages: - Newer technology - Complex algorithms/quantification - Measurements can take several minutes - Optics often misalign due to vibration Infrared (IR) technology is based on the principle of absorbance whereby as the IR light source passes through a sample, the light absorbance is compared to a known standard.
Light/Radiation Based	Raman	Specifications: - Wavelengths 50 to 4,000 cm^{-1} - Detection limit of 0.1% - Difficult to detect materials that are highly colored - Highly effective in water based solutions Advantages: - Relatively fast - Non-destructive for sample - Flexibility in sample size - Minimal to no sample prep required - High specificity - Short measurement times (seconds) Disadvantages: - Newer technology - Complex algorithms/quantification - High cost to maintain - Variability in sensitivity - Background signal interference with measurement Raman technology measures the frequency that scattered photons shift from the light source. This is commonly known as the “Raman shift.” Over the past few years, Raman has been marketed by different companies as an excellent technology to confirm identity for incoming raw materials. Some technologies can be used on the shop floor where materials are stored.

Table 7.1: Commonly Used Types of Analyzers, Basic Capabilities, Limitations, and Highlights (continued)

Sensor Family Group	Analyzer Technology	Nomenclature and Description
Light/Radiation Based	Ultraviolet/Visible (UV/VIS)	Specifications: - Wavelengths 1 to 400 nm* (Visible light = 400 to 800 nm) - Can measure transition metals Advantages: - Relatively fast - Sampling and sample prep not typically performed - Non-destructive for sample - Useful in vessel measurement (e.g., metal tanks) Disadvantages: - Can have complex algorithms/quantification - Difficulties in calibration for quantitative spectroscopic analysis UV/VIS is typically applied in industry in use of tank vessels and cleaning applications. The ability to measure in areas where metal is present and not be impeded is a significant advantage for this technology.
None	Mass Spectroscopy (MS)	Specifications: - Atomic Mass Units (AMUs) are measured Advantages: - Relatively fast - Sampling and sample preparation not typically performed - Can identify substances in a gas-phase sample Disadvantages: - Newer technology - Complex algorithms/quantification - Destructive Measure to sample - True quantification is difficult
Light/Radiation Based	Focused Beam Reflectance Measurement (FBRM)	Specifications: - Utilizes back scatter method of detection measurement - Back scatter time related to particle size and shape Advantages: - Can measure in slurries with high solids composition - Eliminates sampling variance contribution to measurement <i>In situ</i> monitoring is robust for crystalline composition Disadvantages: - Newer technology - Sapphire window – costly material FBRM technology is still relatively new technology. Samples are measured by focusing a laser on a continuously moving sample. An actual probe is typically placed/fixed into the sample vessel. The laser light becomes back scattered and readings are taken from the back scattering light.

Table 7.1: Commonly Used Types of Analyzers, Basic Capabilities, Limitations, and Highlights (continued)

Sensor Family Group	Analyzer Technology	Nomenclature and Description
None	Gas Chromatography (GC)	Specifications: - Sample vaporized and injected onto the head of a chromatographic column Advantages: - Can measure gas or liquid samples - Amount of analyte can be quantified at peak height - Highly sensitive, selective and precise Disadvantages: - Complex algorithms/quantification - Destructive measure to sample - Complex sampling system - Modest time for test results - Cannot directly analyze a sample during processing/manufacturing
None	High Performance Liquid Chromatography (HPLC)	Specifications: - Separates sample by retention time in column Advantages: - Measures liquid samples - Amount of analyte can be quantified at peak height - Highly sensitive, selective and precise Disadvantages: - Complex algorithms/quantification - Lengthy time for test results - Destructive measure to sample HPLC Systems are not typically used online. There has been technology developed in the past few years to allow HPLC to become more portable, but it still very much remains a laboratory platform technology.
Soft Sensors (also called Software Sensors)	I.e., algorithms	Allow prediction of a variable of interest on the basis of already measured variables

Appendix 4

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8 Appendix 4 – References

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Appendix 5

Glossary

9 Appendix 5 – Glossary

9.1 Acronyms

AMU	Atomic Mass Unit
APR	Annual Product Review
CAPA	Corrective Action and Preventative Action
CMC	Chemistry and Manufacturing and Control
CPP	Critical Process Parameter
CQA	Critical Quality Attribute
CUSUM	Cumulative Sum Control Chart
DoE	Design of Experiments
EPC	Engineering Process Control
EWMA	Exponentially-Weighted Moving Average
FBRM	Focused Beam Reflectance Measurement
FCA	Field Corrective Action
FMEA	Failure Mode and Effects Analysis
FRS	Functional Requirement Specification
FTIR	Fourier Transform Infrared
GC	Gas Chromatography
GMP	Good Manufacturing Practice
HPLC	High Performance Liquid Chromatography
IR	Infrared
LCL	Lower (Process) Control Limit
LIF	Laser-Induced Fluorescence
LSL	Lower Specification Limit
MA	Material Attribute
MS	Mass Spectroscopy

MVDA	Multivariate Data Analysis
NIR	Near Infrared
OOS	Out of Specification
OOT	Out Of Trend
OQ	Operational Qualification
PAR	Proven Acceptable Range
PAT	Process Analytical Technology
PDO	Product Design Owner
PFD	Process Flow Diagram
PLC	Programmable Logic Controller
PP&PQMS	Process Performance and Product Quality Monitoring System
PQ	Performance Qualification
PQS	Pharmaceutical Quality System
PSCT	Product Supply Chain Team
QbD	Quality by Design
QRM	Quality Risk Management
QS	Quality Steward
QU	Quality Unit
RTRT	Real Time Release Testing
SME	Subject Matter Expert
SPC	Statistical Process Control
SQC	Statistical Quality Control
UCL	Upper (Process) Control Limit
UDU	Uniformity of Dosage Unit
URS	User Requirement Specification
USL	Upper Specification Limit
UV/VIS	Ultraviolet/Visible

9.2 Definitions

Alert Limit

A limit reached prior to an action limit which, if exceeded, provides an early warning of a potential change from normal operating conditions

Action Limit

A limit which, if exceeded, indicates a significant change from normal conditions.

Capability of a Process (ISO 9000:2005)

Ability of a process to realize a product that will fulfill the requirements of that product. The concept of process capability can also be defined in statistical terms.

Change Management (ICH Q10)

A systematic approach to proposing, evaluating, approving, implementing and reviewing changes.

Continual Improvement (ISO 9000:2005)

Recurring activity to increase the ability to fulfill requirements.

Continuous Process Verification (ICH (Q8(R2)))

An alternative approach to process validation in which manufacturing process performance is continuously monitored and evaluated.

Control Strategy (ICH Q10)

A planned set of controls, derived from current product and process understanding that ensures process performance and product quality. The controls can include parameters and attributes related to drug substance and drug product materials and components, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring or control.

Corrective Action (ISO 9000:2005)

Action to eliminate the cause of a detected non-conformity or other undesirable situation. Note: corrective action is taken to prevent recurrence whereas preventive action is taken to prevent occurrence.

Critical Process Parameter (CPP) (ICH Q8(R2))

A process parameter whose variability has an impact on a critical quality attribute and therefore should be monitored or controlled to ensure the process produces desired quality.

Critical Quality Attribute (CQA) (ICH Q8(R2))

A physical, chemical, biological or microbiological property or characteristic that should be within an appropriate limit, range, or distribution to ensure product quality.

Design Space (ICH Q8(R2))

The multidimensional combination and interaction of input variables (e.g., material attributes) and process parameters that have been demonstrated to provide assurance of quality. Working within the design space is not considered as a change. Movement out of the design space is considered to be a change and would normally initiate a regulatory post approval change process. Design space is proposed by the applicant and is subject to regulatory assessment and approval.

Detectability (ICH Q9)

The ability to discover or determine the existence, presence, or fact of a hazard.

Feedback/Feedforward (ICH Q10)

- Feedback: the modification or control of a process or system by its results or effects.
- Feedforward: the modification or control of a process using its anticipated results or effects. (Oxford Dictionary of English. Oxford University Press; 2003)

Feedback/feedforward can be applied technically in process control strategies and conceptually in quality management. (ICH Q10)

Formal Experimental Design

A structured, organized method for determining the relationship between factors affecting a process and the output of that process. Also known as “Design of Experiments.”

Innovation (ICH Q10)

The introduction of new technologies or methodologies.

Knowledge Management (ICH Q10)

Systematic approach to acquiring, analyzing, storing, and disseminating information related to products, manufacturing processes and components.

Lifecycle (ICH Q8(R2))

All phases in the life of a product from the initial development through marketing until the product's discontinuation.

Outsourced Activities (ICH Q10)

Activities conducted by a contract acceptor under a written agreement with a contract giver.

Performance Indicators (ICH Q10)

Measurable values used to quantify quality objectives to reflect the performance of an organization, process, or system, also known as “performance metrics” in some regions.

Pharmaceutical Quality System (PQS) (ICH Q10 based upon ISO 9000:2005)

Management system to direct and control a pharmaceutical company with regard to quality.

Preventive Action (ISO 9000:2005)

Action to eliminate the cause of a potential non-conformity or other undesirable potential situation. Note: preventive action is taken to prevent occurrence whereas corrective action is taken to prevent recurrence.

Preventative Action and Corrective Action System (ICH Q10)

A system for implementing corrective actions and preventative actions resulting from an investigation of complaints, product rejections, non-conformances, recalls, deviations, audits, regulatory inspections and findings, and trends from process performance and product quality monitoring.

Process Analytical Technology (PAT) (ICH Q8(R2))

A system for designing, analyzing, and controlling manufacturing through timely measurements (i.e., during processing) of critical quality and performance attributes of raw and in-process materials and processes with the goal of ensuring final product quality.

Process Parameter

A process variable (e.g., temperature, compression force) that can be assigned values to be used as control levels or operating limits.

Process Performance and Product Quality Monitoring System (PP&PQMS) (ICH Q10)

A system for the monitoring of process performance and product quality to ensure a state of control is maintained.

Process Robustness (ICH Q8(R2))

Ability of a process to tolerate variability of materials and changes of the process and equipment without negative impact on quality.

Product Realization (ICH Q10)

Achievement of a product with the quality attributes appropriate to meet the needs of patients, health care professionals, and regulatory authorities (including compliance with marketing authorization) and internal customers requirements.

Proven Acceptable Range (ICH Q8(R2))

A characterized range of a process parameter for which operation within this range, while keeping other parameters constant, will result in producing a material meeting relevant quality criteria.

Quality (ICH Q6A)

The suitability of either a drug substance or drug product for its intended use. This term includes such attributes as the identity, strength, and purity (from ICH Q6A Specifications: Test Procedures and Acceptance Criteria for New Drug Substances and New Drug Products: Chemical Substances).

Quality Attribute

A physical, chemical, or microbiological property or characteristic that directly or indirectly relates to pre-defined product quality (safety, identity, strength, purity, and marketability of the product).

Quality by Design (QbD) (ICH Q8(R2))

A systematic approach to development that begins with predefined objectives and emphasizes product and process understanding and process control, based on sound science and quality risk management.

Quality Risk Management (ICH Q9)

A systematic process for the assessment, control, communication, and review of risks to the quality of the drug (medicinal) product across the product lifecycle.

Quality Target Product Profile (ICH Q8(R2))

A prospective summary of the quality characteristics of a drug product that ideally will be achieved to ensure the desired quality, taking into account safety and efficacy of the drug product.

Quality Risk Management (ICH Q9)

A systematic process for the assessment, control, communication and review of risks to the quality of the drug (medicinal) product across the product lifecycle.

Quality System (ICH Q9)

The sum of all aspects of a system that implements quality policy and ensures that quality objectives are met.

Risk (ISO/IEC Guide 51)

The combination of the probability of occurrence of harm and the severity of that harm.

Risk Acceptance (ISO Guide 73)

The decision to accept risk.

Risk Analysis (ICH Q9)

The estimation of the risk associated with the identified hazards.

Risk Assessment (ICH Q9)

A systematic process of organizing information to support a risk decision to be made within a risk management process. It consists of the identification of hazards and the analysis and evaluation of risks associated with exposure to those hazards.

Risk Communication (ICH Q9)

The sharing of information about risk and risk management between the decision maker and other stakeholders.

Risk Control (ISO Guide 73)

Actions implementing risk management decisions.



**Connecting a World of
Pharmaceutical Knowledge**

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